

Determination of a metric and a unitary connection from one boundary patch

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October 5, 2026

Abstract

We prove that zero-frequency boundary measurements on any nonempty open boundary patch determine both a smooth Riemannian metric and a smooth unitary connection on the trivial Hermitian rank-two bundle over a compact connected smooth manifold of dimension at least three with smooth boundary. Both inputs and observations are restricted to the same patch. The metric and connection are determined up to a diffeomorphism and a unitary gauge that restrict to the identity on the measured patch.

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1 Introduction

A connection Laplacian couples the geometry of a manifold to parallel transport in a vector bundle. Its boundary measurements therefore pose two simultaneous inverse problems: determining the metric that governs its principal part and determining the connection that governs the transport of vector-valued solutions. We consider both unknowns at zero frequency, with all measurements confined to one boundary patch.

Let M be a compact connected smooth manifold of dimension $n \geq 3$, with nonempty smooth boundary, and let $\Gamma \subset \partial M$ be a nonempty relatively open proper subset. The bundle is the trivial Hermitian bundle $M \times \mathbb{C}^2$. A smooth unitary connection is written $d_A = d + A$, where

$$A \in C^\infty(M; T^*M \otimes \mathfrak{u}(2)), \quad \mathfrak{u}(2) = \{B \in \mathbb{C}^{2 \times 2} : B^* = -B\}.$$

For a smooth Riemannian metric g , write $L_{g,A} = d_A^{*g} d_A$. Its zero-Dirichlet realization is invertible: a section of zero energy is parallel, and a parallel section with zero boundary trace is zero. Elliptic coercivity then gives, for each $f \in C_c^\infty(\Gamma; \mathbb{C}^2)$ extended by zero to ∂M , a unique solution $u_f^{g,A}$ of

$$L_{g,A} u_f^{g,A} = 0 \quad \text{in } M, \quad u_f^{g,A}|_{\partial M} = f.$$

The measured object is the sesquilinear energy form

$$\langle \Lambda_{g,A,\Gamma} f, h \rangle = \int_M \langle d_A u_f^{g,A}, d_A u_h^{g,A} \rangle_g dV_g, \quad f, h \in C_c^\infty(\Gamma; \mathbb{C}^2). \quad (1.1)$$

The Hermitian product is linear in its first argument. All integrations use densities; no orientation of M is required. In particular, (1.1) specifies the boundary measurement without presupposing a boundary metric or a boundary measure.

Theorem 1.1. *Let M and Γ be as above. For $j = 1, 2$, let g_j be a smooth Riemannian metric on M and let $A_j \in C^\infty(M; T^*M \otimes \mathfrak{u}(2))$. If*

$$\langle \Lambda_{g_1, A_1, \Gamma} f, h \rangle = \langle \Lambda_{g_2, A_2, \Gamma} f, h \rangle \quad \text{for all } f, h \in C_c^\infty(\Gamma; \mathbb{C}^2),$$

then there are a smooth diffeomorphism $\Phi : M \rightarrow M$ and a smooth map $U : M \rightarrow U(2)$ such that

$$\Phi|_\Gamma = \text{Id}, \quad U|_\Gamma = I, \quad g_2 = \Phi^* g_1, \quad A_2 = U^{-1}(\Phi^* A_1)U + U^{-1}dU. \quad (1.2)$$

Here I denotes the 2×2 identity matrix.

The two transformations in (1.2) preserve (1.1): the corresponding change of solution is $u_2 = U^{-1}(u_1 \circ \Phi)$. Thus the conclusion describes precisely the ambiguities inherent in these measurements. The patch need not be connected, and the manifold may have several boundary components.

1.1 Context and contribution

Calderón’s inverse conductivity problem asks whether electrical measurements at the boundary determine an interior conductivity [3]. For smooth isotropic conductivities in dimension at least three, global uniqueness was established by Sylvester and Uhlmann [22]. In the anisotropic formulation the unknown is a metric, or equivalently a positive conductivity tensor density, and boundary-fixing diffeomorphisms give an unavoidable invariance. Lee and Uhlmann proved a boundary determination result and uniqueness in the real-analytic setting under geometric and topological hypotheses [16]. Lassas and Uhlmann subsequently recovered real-analytic manifolds from local boundary information [14]. The Poisson embedding approach of Lassas, Liimatainen, and Salo [15] expresses the geometry through the values of harmonic solutions. We use the same broad principle, with source solutions of a connection Laplacian providing both points and fiber identifications.

For connections, Albin, Guillarmou, Tzou, and Uhlmann recovered Hermitian connections and matrix potentials from full Cauchy data on fixed surfaces [1]. In higher dimensions, Cekić established recovery for line bundles on certain known conformally transversally anisotropic geometries [4], and for smooth unitary Yang–Mills connections of arbitrary rank from same-patch measurements on an arbitrary fixed smooth metric [5, Theorem 1.3]. His complex parallel transport method also recovers arbitrary-rank connections and matrix potentials from full boundary data on fixed geometries conformal to subdomains of $\mathbb{R}^2 \times M_0$ [6, Theorem 1.1]. In the real-analytic category, Gabdurakhmanov and Kokarev proved joint recovery of metric, Euclidean bundle, and compatible connection from a boundary patch [9, Theorem 1.1]. Their use of Green kernels to identify the base and fibers is a close geometric antecedent. Theorem 1.1 concerns an unknown smooth metric and an arbitrary smooth unitary connection on the fixed trivial rank-two bundle; it imposes no Yang–Mills equation on that connection.

The scalar companion [18] develops a smooth continuation argument based on separately harmonic Green kernels and shrinking geodesic spheres. The present proof uses its product continuation, sphere geometry, and scalar angular estimate [18, Sections 3–5]. We give the analytic arguments needed here, including their extension to systems with scalar principal part. The scalar uniqueness theorem itself is not applied to the matrix-valued boundary data.

The additional difficulty appears in the first moments of the transfer between local solution spaces. In scalar harmonic coordinates those moments can be normalized away. For a connection, a harmonic frame removes the zeroth-order term, but the transfer still has a matrix-valued first moment. We represent it by matrices $D^1, \dots, D^n$. Their components complementary to the real scalar matrices satisfy a first-order system whose principal symbol is the conformal Killing symbol. In dimension at least three this symbol is of finite type: two prolongations determine all third derivatives from lower jets. The real scalar part of the displacement is fixed by the coordinate normalization. Together these facts turn an initial square-root estimate on D into the stronger estimate needed to continue the metric and connection across a frontier point. This reduction is the main matrix-specific step of the proof.

1.2 How the proof is organized

Boundary symbol factorization first determines all metric and connection jets on Γ , after imposing normal gauge. Attaching a small exterior cap then converts the boundary form into equality of Green matrices on an open interior set. Source solutions supported in that set separate points, span fiber values, and realize the allowed harmonic jets. They define a

maximal local isometry with a unitary bundle identification; Section 2 prepares coordinates and harmonic frames at any putative frontier point.

The local continuation mechanism has three stages. First, Section 3 continues a mixed Green matrix, initially known when either variable lies in the matched region. Two quantitative unique-continuation estimates, one in each variable, allow continuation across suitable hypersurfaces in the product. Second, Sections 4 and 5 use this kernel to transfer harmonic functions across small moving spheres. The transfer is represented by a matrix density of total mass I . An angular coercivity estimate bounds that density uniformly even as the spheres shrink. The matrix lower-order terms are absorbed in the same estimate.

Finally, Section 6 extracts the first and second moments of the transfer. The finite-type argument described above improves the difference of the normalized inverse metrics from order r to order $r^{3/2}$, where r is the sphere radius. A continuity argument then lets the radii collapse, proving local agreement of the normalized operators. Section 7 removes the remaining conformal factor and extends the resulting isometry and unitary identification over the whole manifold, including every component of the measured patch.

2 Boundary data, exterior sources, and contact coordinates

We first replace the measurements by a family of interior source solutions. The boundary symbol determines jets of both the metric and the connection, after a boundary-identity change of gauge. These jets allow a common exterior cap, on which the Green matrices agree. Source evaluations in that cap then identify points and fibers simultaneously. The final part of the section prepares coordinates and harmonic frames at a possible frontier of this identification. The cap and source-evaluation construction follows the scalar construction in [18, Section 2]; we give the changes needed for unitary connections and matrix-valued evaluations.

2.1 Dirichlet solutions and boundary jets

We use positive metric densities throughout. Write $L_{g,A} = d_A^{*g} d_A$, with positive principal symbol, and let $L_{g,A,D}$ denote its zero-Dirichlet realization. Its quadratic form is coercive on $H_0^1(M; \mathbb{C}^2)$. Indeed, the usual elliptic form estimate and compactness reduce coercivity to the absence of a kernel. A kernel element satisfies $d_A u = 0$; its norm is constant, and its zero boundary trace makes it zero. Smooth sources and boundary data therefore have unique solutions smooth up to the boundary. The same facts hold on every smooth extension used below and on sufficiently small coordinate balls; we use the standard smooth elliptic Dirichlet theory of [23, Chapter 5].

We also use open-set unique continuation for systems with a smooth, real scalar elliptic principal part and smooth matrix lower-order terms; see [2, Remark 3]. The usual scalar local Carleman estimate, summed over components, proves this statement: after conjugation by the scalar weight, a matrix first-order term is bounded by a constant times $\|\nabla v\| + \tau\|v\|$, and is absorbed for large Carleman parameter τ . See [11, Chapter XXVIII] for the underlying scalar estimates. Boundary Cauchy uniqueness follows from the same interior statement. Extend the coefficients smoothly across a boundary patch and extend a solution with zero trace and zero covariant normal derivative by zero. Green's formula leaves no boundary distribution, so the extension solves the equation weakly. Interior regularity and unique continuation then give vanishing near the patch and throughout the connected interior.

A unitary gauge $V : M \rightarrow U(2)$ changes the connection to

$$A^V = V^{-1} A V + V^{-1} dV, \quad d_{A^V}(V^{-1} u) = V^{-1} d_A u. \quad (2.1)$$

A gauge equal to I on the boundary preserves the measured form. In inward boundary-normal coordinates (z, s) , solve $\partial_s V = -A_s V$, $V(z, 0) = I$. This gives $A_s^V = 0$. The gauge can be made global: in a collar replace the time argument of this solution by a smooth function equal to s in a smaller collar and equal to zero near the inner edge, and set $V = I$ farther inside. The ODE preserves unitarity. Apply this construction separately to A_1, A_2 , using gauges V_1, V_2 , and retain their names for the final return to the original trivializations. Until then, A_j denotes the gauged connection.

Lemma 2.1 (Boundary jets). *Suppose the local energy forms of (g_1, A_1) and (g_2, A_2) agree on Γ . In their respective boundary-normal coordinates, based on the same boundary coordinates, and in the normal gauges just constructed, the full Taylor jets of g_1, A_1 and g_2, A_2 agree at every point of Γ .*

Proof. Write $g = ds^2 + k(s, z)$, with k a tangential positive matrix, and put $\rho = (k^{ab}\xi_a\xi_b)^{1/2}$. Green's formula identifies the measured operator with the density $\nabla_\nu^A u dS_g$. Relative to the coordinate density, its principal symbol is

$$(\det k)^{1/2} \rho I.$$

Its square determines $C = (\det k)k^{-1}$. Since the tangential dimension is $m = n - 1$,

$$\det C = (\det k)^{m-1}, \quad k = (\det C)^{1/(m-1)} C^{-1}.$$

Thus the boundary metrics and densities agree. Division by their common density gives equality of the ordinary covariant unit-normal DN operators locally on Γ . Localization on both sides by cutoffs supported in Γ gives equality of their full symbols.

We spell out the symbol calculation, extending the metric factorization of [16] to the present scalar-principal system. Recovery of a connection together with the metric by this even-odd symbol separation is also developed in [8, Theorem 1.1 and Section 3.3]; the calculation below uses complex Hermitian fibers. In normal gauge,

$$L_{g,A} = -\partial_s^2 - \alpha \partial_s + P, \quad \alpha = \frac{1}{2} \operatorname{tr}(k^{-1} \partial_s k),$$

where P is the connection Laplacian on each tangential slice. With $D_z = -i\partial_z$, its degree-one symbol consists of the scalar metric term and $-2ik^{ab}A_a\xi_b$. Its degree-zero symbol uses only slice coefficients and their tangential derivatives. The local Dirichlet factorization has the form

$$L_{g,A} = (-\partial_s + B - \alpha)(\partial_s + B) \pmod{\Psi^{-\infty}}, \quad b \# b - \alpha b - \partial_s b \sim \sigma(P),$$

where $b \sim \rho I + b_0 + b_{-1} + \dots$ is a left symbol and

$$b \# c \sim \sum_{\mu} \frac{(1/i)^{|\mu|}}{\mu!} (\partial_\xi^\mu b)(\partial_z^\mu c).$$

The positive root ρI selects the decaying Dirichlet branch. The Poisson parametrix consequently identifies $B(0)$, modulo a smoothing operator, with the outward normal derivative. This factorization is valid for smooth coefficients: its principal root is scalar and each subsequent matrix coefficient is solved by scalar division by 2ρ . The true inverse Dirichlet problem changes only the local smoothing remainder.

The part of b_0 involving the new jets is

$$\frac{1}{4} \left(\operatorname{tr}_k \partial_s k - \frac{(\partial_s k)(\xi^\sharp, \xi^\sharp)}{\rho^2} \right) I - \frac{ik^{ab}A_a\xi_b}{\rho}, \quad \xi^\sharp = k^{-1}\xi. \quad (2.2)$$

This follows from $\partial_s \rho = -(\partial_s k)(\xi^\sharp, \xi^\sharp)/(2\rho)$ and the degree-one factorization equation. At degree $1 - j$, $j \geq 2$, the new normal derivatives arise only in $\partial_s b_{2-j}/(2\rho)$. Inductively their contribution is

$$\frac{1}{(2\rho)^{j-1}} \left[\frac{1}{4} \left(\text{tr}_k \partial_s^j k - \frac{(\partial_s^j k)(\xi^\sharp, \xi^\sharp)}{\rho^2} \right) I - \frac{ik^{ab}(\partial_s^{j-1} A_a)\xi_b}{\rho} \right]. \quad (2.3)$$

Every omitted term involves lower normal jets, with tangential derivatives. Differentiating ρ , a raised index, or a coefficient multiplying the preceding highest jet produces only such lower-jet expressions.

Subtract the two symbol recursions after their lower jets have been identified. The metric term in (2.3) is even in ξ , whereas the connection term is odd. The odd term determines all components of the new connection jet. If the even term vanishes for a symmetric tensor T , then $T(v, v) = \text{tr}_k(T)$ for every unit vector v . Polarization gives $T = \text{tr}_k(T)k$, and taking trace gives $(m-1)\text{tr}_k(T) = 0$. As $m \geq 2$, $T = 0$. Beginning with the recovered boundary metric, this proves all normal jets by induction; their tangential derivatives recover all mixed jets. The remaining normal metric components and normal connection component are fixed by the coordinate and gauge choices. $\square$

2.2 A common cap and its Green matrices

Jet equality does not assert equality in an interior collar. It provides instead a common smooth extension on the exterior side of a smaller patch. The following variational argument turns that exterior region into common interior data.

Proposition 2.2 (Common exterior sources). *There are compact connected smooth extensions N_j of the original manifold, with nonempty smooth boundary, and a common connected open exterior cap $E \subset N_j^\circ$, attached through a nonempty open patch $P \Subset \Gamma$, such that the extended metrics and connections agree on E . For each $f \in C_c^\infty(E; \mathbb{C}^2)$, the solutions $L_{j,D}^{-1}f$ agree on E . The Dirichlet Green matrices, normalized using metric volume, satisfy*

$$G_1(x, y) = G_2(x, y) \quad (x, y \in E, x \neq y), \quad G_j(x, y) = G_j(y, x)^*. \quad (2.4)$$

Proof. Choose a boundary chart compactly contained in Γ , and a nonnegative smooth bump b whose support is compact in that chart and whose positivity set $P = \{b > 0\}$ is nonempty and connected. Using the separate normal collars, move the boundary from $s = 0$ to the graph $s = -b(z)$, with b small enough to remain inside an extended collar. The common open cap is

$$E = \{(z, s) : -b(z) < s < 0\}.$$

Extend the first coefficient fields smoothly to negative s , retaining positive definiteness of the metric and skew-Hermitian values of the connection. Use the same exterior extension for the second pair. Lemma 2.1 makes the resulting pasting smooth to all orders on a neighborhood of $\text{supp } b$. Restricting it to $s \geq -b(z)$ gives smoothness also at the flat edges of the bump. The bundles remain trivial in the extended normal gauges.

For the energy comparison only, identify the extensions by the identity on the original M and by cap coordinates on E . In collar charts this map and its inverse are continuous and piecewise smooth with bounded first derivatives, and hence bi-Lipschitz. They identify the relevant H_0^1 spaces. Fix a vector source f compactly supported in E . The solution $w_j = L_{j,D}^{-1}f$ is the unique minimizer of

$$\mathcal{J}_j(w) = \frac{1}{2} \int_{N_j} |d_{A_j} w|_{g_j}^2 dV_{g_j} - \text{Re} \int_E \langle f, w \rangle dV_{g_j} \quad \text{on } H_0^1(N_j; \mathbb{C}^2).$$

Its trace on the original boundary is smooth and supported in $\text{supp } b \in \Gamma$, because the original and extended boundaries coincide wherever $b = 0$. Transport w_1 to N_2 , and inside the original M replace it by the second harmonic extension of this trace. The replacing difference has zero trace on ∂M , so its zero extension is in $H_0^1(N_2)$. The competitor is therefore admissible even at the narrowing edges of the cap.

On M , the source is zero and w_1 is the first harmonic extension of its trace. Equality of the measured quadratic forms makes its interior energy equal to that of the replacement. On E , both coefficients and the source pairing are unchanged. Thus $\min \mathcal{J}_2 \leq \min \mathcal{J}_1$. Reversing the argument gives equality, and uniqueness of the minimizer proves $w_1 = w_2$ on E .

Normalize the Green matrix by

$$L_{j,x}G_j(x,y) = \delta_y^{g_j}(x)I, \quad G_j(\cdot, y)|_{\partial N_j} = 0, \quad L_{j,D}^{-1}f(x) = \int_{N_j} G_j(x,y)f(y) dV_{g_j}(y).$$

The source densities on E coincide. Varying the source gives equality of the distribution kernels on $E \times E$, hence pointwise equality off the diagonal. Self-adjointness of the Dirichlet inverse gives the adjoint symmetry in (2.4). $\square$

2.3 Evaluations identify points and fibers

Fix a nonempty open set $U_0 \Subset E$. For $f \in C_c^\infty(U_0; \mathbb{C}^2)$, set

$$w_{j,f} = L_{j,D}^{-1}f, \quad I_j(p)f = w_{j,f}(p). \quad (2.5)$$

Thus $I_j(p)$ is a linear map from the common source space to the fiber at p . The maps agree on E . This construction is related to the source and Poisson embeddings of [15, Section 6.1 and Appendix A]. The following point-distribution proof, in the spirit of [17, Chapter III, Section 3], supplies the precise system-valued separation needed here.

Lemma 2.3 (Values and harmonic two-jets). *Let N be a compact connected smooth manifold of dimension $n \geq 2$ with nonempty boundary, let $L = d_A^{*g}d_A$, and let $U \Subset N^\circ$ be a nonempty open source set.*

1. *At each interior point, $f \mapsto L_D^{-1}f(p)$ is onto $\mathbb{C}^2$. At any two distinct interior points the two values are jointly onto $\mathbb{C}^2 \oplus \mathbb{C}^2$.*
2. *If $p \notin \bar{U}$, the second jets of $L_D^{-1}f$ at p fill exactly the linear space of jets satisfying $Lu(p) = 0$.*
3. *Each evaluation is zero on ∂N , and for fixed f varies continuously up to that boundary.*

Proof. The last assertion is smooth Dirichlet regularity. For the first two, a complex-linear functional on values or jets is represented by a dual-vector distribution T , supported at one or two points. Suppose it annihilates all source solutions, and define $v = (L_D^{-1})^t T$ by distributional transposition. Then

$$\langle v, f \rangle = \langle T, L_D^{-1}f \rangle = 0 \quad (f \in C_c^\infty(U; \mathbb{C}^2)), \quad L^t v = T.$$

Here the transpose and pairing are bilinear on the dual bundle; no complex conjugation convention is needed. Elliptic regularity makes v smooth off the support of T . The interior with one or two points removed is connected: a path can be diverted around each point in a punctured coordinate ball when $n \geq 2$. Unique continuation from the nonempty set

obtained by removing these points from U makes v supported at those points in the interior. If a point belongs to U , the asserted initial vanishing is distributional there as well.

At each support point a distribution is a finite sum of derivatives of the point mass. If its highest derivative order is k , its image under L^t has exact highest order $k + 2$. Indeed, the highest homogeneous coefficient vector polynomial is multiplied by the scalar elliptic quadratic polynomial; this multiplication is injective. Consequently no nonzero order-zero distribution can be the image of a point-supported distribution. There are therefore no value annihilators, proving both surjectivity claims.

For a two-jet annihilator at $p \notin \bar{U}$, the image T has order at most two. Thus $v = c\delta_p$, with c a dual-fiber vector. Its image is precisely the functional $u \mapsto c(Lu(p))$. Conversely all such functionals annihilate the source solutions. Finite-dimensional duality gives the exact stated jet space. In particular, values and gradients may be prescribed freely, since the Hessian trace can be chosen to satisfy the equation. $\square$

We can now define the correspondence whose continuation will prove the theorem. A *match* consists of a local isometry $\Phi : W \rightarrow N_1^\circ$, with $W \subset N_2^\circ$ connected, open, and containing E , and a smooth unitary bundle map J from the second fiber at p to the first fiber at $\Phi(p)$. We require

$$\Phi|_E = \text{Id}, \quad J|_E = I, \quad d_{\Phi^*A_1}(Ju) = Jd_{A_2}u, \quad J(p)I_2(p) = I_1(\Phi(p)). \quad (2.6)$$

In the connection identity the first connection is pulled back to W . The identity on E is a match.

Lemma 2.4 (The maximal match). *All matches agree on overlaps and are injective on base points. Their union is a unique maximal match (Φ, J) on a connected open set $W \subset N_2^\circ$. On this set,*

$$G_1(\Phi(p), \Phi(q)) = J(p)G_2(p, q)J(q)^* \quad (p, q \in W, p \neq q). \quad (2.7)$$

Proof. If two possible images of the same point were distinct, their evaluation identities would impose a fixed invertible linear relation between the evaluations at two distinct points of N_1° . This contradicts joint surjectivity in Lemma 2.3. Once the image agrees, surjectivity at the source point determines J uniquely. The same argument using the evaluations in N_2 rules out two base points with the same image. Thus the maps glue on every overlap. Their union is connected because every domain contains E , and retains all the properties in (2.6).

For fixed p , the evaluation identity and the common source density give (2.7) when $q \in U_0$, away from the pole. As a function of q , the adjoint of each side solves the same pulled-back connection equation after the bundle identification. Injectivity of Φ ensures that $q = p$ is the only possible pole. Unique continuation on the connected set $W \setminus \{p\}$ proves the identity everywhere claimed. $\square$

2.4 Coordinates at a possible interior frontier

It remains to show that the maximal domain is the entire interior. We next prepare the local data needed to enlarge it. The use of harmonic *frames* is essential: constant coefficient columns in these frames solve the equation, even when the connection has no nonzero parallel sections.

Proposition 2.5 (Contact data). *If the maximal domain W has an interior frontier, there are a point $p \in \partial W \cap N_2^\circ$, a point $q \in N_1^\circ$, and charts about p, q , both with coordinate range a neighborhood of $0 \in \mathbb{R}^n$, with the following properties.*

1. Both charts avoid $\overline{U_0}$. The old match is coordinate identity on its domain in these charts. That domain contains a region $x_n < \kappa(x')$, where κ is smooth, $\kappa(0) = 0$, and $d\kappa(0) = 0$.
2. In smooth local unitary frames, the metrics, connection matrices, and Green matrices agree on the known region (for the Green matrices, on its square off the diagonal), and $g_1(0) = g_2(0) = I_n$.
3. There are invertible matrix functions F_j with harmonic columns, chosen as corresponding source solutions. In these harmonic frames there are corresponding harmonic matrix functions X_j^l , $1 \leq l \leq n$, satisfying

$$\frac{1}{2} \operatorname{Re} \operatorname{tr} X_j^l(x) = x^l, \quad X_j^l(0) = 0, \quad \partial_i X_j^l(0) = \delta_i^l I. \quad (2.8)$$

4. There are finitely many further corresponding harmonic column functions with zero value and gradient at 0, whose Hessians at 0 span the space of symmetric $\mathbb{C}^2$ -valued tensors H satisfying $\sum_i H_{ii} = 0$. All these paired functions, including the frames, agree on the known region and have identical jets at 0.

Proof. Choose a small coordinate ball near an interior frontier and a point of W close enough to that frontier that its distance to the complement is attained inside the chart. The Euclidean ball with that center and radius lies in W and touches its complement at an interior point p . Its boundary gives the required smooth one-sided contact hypersurface.

If $p_k \in W$ tends to p , compactness of N_1 and $U(2)$, using the fixed global trivializations, gives a subsequence for which $\Phi(p_k) \rightarrow q$ and $J(p_k) \rightarrow J_0$. Continuity of every source solution gives

$$J_0 I_2(p) = I_1(q). \quad (2.9)$$

The point q cannot lie on the boundary, where the right side would be zero, since $I_2(p)$ is onto. Two distinct possible limits q contradict joint surjectivity on N_1 ; at a fixed q , surjectivity of $I_2(p)$ determines J_0 . The full limits therefore exist. Also $q \notin E$: otherwise its identity match on E and (2.9) would relate the evaluations at the distinct points q, p of N_2 . Both p and q consequently avoid $\overline{U_0}$.

Choose two source solutions whose columns give an invertible matrix $F_1(q)$. Their corresponding second-side columns give $F_2(p)$, invertible by (2.9). Shrink the charts so both matrices are invertible. In a harmonic frame, $v = F_j^{-1}w$ satisfies an equation with scalar elliptic principal part and no zeroth-order term, since every column of F_j is harmonic. Lemma 2.3 is invariant under this smooth frame change. It therefore supplies matrices X_1^l with value zero and gradients equal to any chosen real coordinate covectors times I . Choose those covectors independently, and let X_2^l be the corresponding source matrices expressed in F_2 .

Set $x^l = \frac{1}{2} \operatorname{Re} \operatorname{tr} X_1^l$ and $y^l = \frac{1}{2} \operatorname{Re} \operatorname{tr} X_2^l$. The first functions form coordinates at q . On matched points, the source identity implies $JF_2 = F_1 \circ \Phi$ and $X_2^l = X_1^l \circ \Phi$. Thus the scalar coordinates and the metric Gram matrices of their differentials agree under the isometry. Passing to p, q , their Gram matrices agree and are positive definite. Hence the y^l are coordinates at p . Fix a target chart neighborhood U_1 on which the x^l are coordinates. The full convergence of partner points permits a source chart neighborhood U_2 so small that $\Phi(W \cap U_2) \subset U_1$. Choose a small coordinate ball Ω about zero contained in both $x(U_1)$ and $y(U_2)$, and restrict the two charts to their inverse images of Ω . For a matched point in the restricted source chart, $x(\Phi(p')) = y(p') \in \Omega$; its partner therefore lies in the restricted target chart. Thus the old map is coordinate identity throughout its domain in these charts.

The matrix gradient identity at q now reads $\partial_i X_1^l(0) = \delta_i^l I$; equality on the one-sided region and smoothness give the same identity for X_2^l .

Because the harmonic-frame equation has no zeroth-order term, a jet with zero value and gradient is constrained only by $g_1^{ij}(0)H_{ij} = 0$. The jet lemma supplies a finite basis of these Hessians and corresponding source pairs. They agree on the known region, so their complete jets agree at the contact point. The same conclusion holds for every smooth coefficient or paired function already identified there.

Finally polar-orthonormalize the original harmonic frames: write $F_j = H_j S_j$, with H_j unitary and S_j positive Hermitian. On the matched set, $JF_2 = F_1$ implies $S_2 = S_1$ and $JH_2 = H_1$. In the unitary frames H_j , the old bundle identification is therefore I ; connection matrices and Green matrices agree there. We continue to write F_j for the harmonic-frame matrices in these unitary frames. A common real linear coordinate change makes the metric at zero Euclidean and the contact hyperplane horizontal. Apply the same linear change to the list of X_j^l ; this preserves all identities in (2.8). Reverse the final normal coordinate if necessary to put the known side below its graph. In these coordinates the Hessian constraint is the one in the statement. $\square$

2.5 Dilation of the contact data

The local analysis uses the preceding charts on a small physical scale and a fixed coordinate range. For $\lambda > 0$, set

$$\begin{aligned} g_{j,\lambda}(x) &= g_j(\lambda x), & A_{j,\lambda}(x) &= \lambda A_j(\lambda x), \\ G_{j,\lambda}(x, y) &= \lambda^{n-2} G_j(\lambda x, \lambda y), & F_{j,\lambda}(x) &= F_j(\lambda x). \end{aligned} \tag{2.10}$$

Here the metric normalization is λ^{-2} times the pullback metric. Thus the pulled-back connection Laplacian is multiplied by λ^2 , and the factor λ^{n-2} in the kernel exactly compensates for metric volume. In particular the displayed kernel still has unit matrix point source for $g_{j,\lambda}$. Harmonic functions and the harmonic frames dilate by composition, without a multiplicative factor.

On every fixed bounded coordinate range, these metrics converge smoothly to I_n , the connection matrices converge smoothly to zero, and all fixed finite orders of their coefficients and frames remain bounded. Source-free harmonic-frame operators have zero zeroth-order terms; their ordinary-coordinate first-order coefficients tend to zero. The known side becomes $x_n < \lambda^{-1} \kappa(\lambda x')$, which contains every fixed bounded region lying strictly below $x_n = 0$ for all sufficiently small λ .

We also have uniform kernel bounds on fixed separated sets. An interior parametrix of order -2 for the original smooth scalar-principal system gives $|G_j(x, y)| \leq C|x - y|^{2-n}$ in a sufficiently small fixed chart, with a smooth remainder; interior estimates give the corresponding bounds for derivatives away from the diagonal. After (2.10), every fixed positive lower bound for $|x - y|$ therefore gives bounds of any prescribed finite order, uniformly for small λ . The local constructions below first specify their bounded ranges and positive separation margins and then restrict λ . They never demand a bound uniform as the separation tends to zero from this initialization alone.

For the local argument, our convention for a normalized equation in a harmonic frame is

$$\mathcal{L}_j v = -c_j F_j^{-1} L_{g_j, A_j}(F_j v), \quad c_j > 0. \tag{2.11}$$

The scalar multiplier acts outside the differential operator. Its second-derivative coefficient matrix is $c_j g_j^{-1}$, and its zeroth-order term is zero. This convention applies to both the original and the dilated data; the functions c_j will be chosen in Section 4.

The remaining local task is to continue the match through zero. Sections 3–6 will prove that the metrics are conformal and that suitable normalizations (2.11) give equal operators on a neighborhood of zero. Section 7 will then use the vanishing zeroth-order terms to remove the conformal factor and prove that $F_1 F_2^{-1}$ is a unitary connection identification. Unique continuation extends every source evaluation, so the local identification enlarges the maximal match.

3 Continuation of solutions in two variables

The local data give a matrix Green kernel when at least one variable lies in the region where the two structures already agree. We must continue this kernel to separated pairs outside that region. This section proves the required continuation for elliptic systems acting on separate tensor indices. The geometric construction and its scalar analytic foundation are those of [18, Section 3]; we reproduce the argument, including the interpolation and overlap constructions, and make the extension to systems explicit. No scalar inverse-problem uniqueness theorem is used.

3.1 Separate equations and extension from a cross

Let V_i be finite-dimensional Hermitian vector spaces. In a coordinate domain in $\mathbb{R}^n$, consider an operator

$$\mathbf{L}_i = -\Delta_{g_i} I_{V_i} + C_i^a \partial_a + D_i,$$

where g_i is a smooth Riemannian metric and C_i^a, D_i are smooth endomorphism-valued coefficients. A *product solution* is a $V_1 \otimes V_2$ -valued function $F(x, y)$ satisfying

$$(\mathbf{L}_1^x \otimes I)F = 0, \quad (I \otimes \mathbf{L}_2^y)F = 0.$$

We suppress the identity factors below. The coefficients of one equation act only on its indicated tensor index and depend only on its indicated variable. In particular, the two equations commute. Their sum has scalar real elliptic principal part on the product; ordinary unique continuation therefore identifies product solutions that agree on a nonempty open subset of a connected common domain. Multiplication of either equation by a positive scalar function does not change its solutions or any continuation constructed from them.

For matrix Green kernels the first tensor factor is the output fiber. The second is the conjugate of the input fiber: the equation in y is obtained by conjugating the coefficients of the second connection Laplacian. Equivalently, the adjoint columns of the kernel solve the second equation. This convention places the two matrix actions on separate indices, as required here.

The first construction extends compatible data on $(D_1 \times S_2) \cup (S_1 \times D_2)$. Its mechanism is a basis that is orthogonal both on a large domain and after restriction to the observed set. Common-basis methods have a classical antecedent in separately harmonic extension [24]; singular systems also enter quantitative Runge approximation [21]. The interpolation threshold and the construction below follow the scalar companion.

Lemma 3.1 (Extension from a cross). *Let D_i be bounded connected coordinate domains, and let $S_i \subseteq D_i$ be nonempty open sets. Suppose a product solution F is given consistently on*

$$(D_1 \times S_2) \cup (S_1 \times D_2),$$

with the sum of its two L^2 norms at most M . Suppose $O_i \subseteq D_i$ are open sets and every V_i -valued solution w of $\mathbf{L}_i w = 0$ on D_i satisfies

$$\|w\|_{L^2(O_i)} \leq C \|w\|_{L^2(S_i)}^{\gamma_i} \|w\|_{L^2(D_i)}^{1-\gamma_i}, \quad 0 < \gamma_i < 1, \quad \gamma_1 + \gamma_2 > 1. \quad (3.1)$$

Then F extends as a product solution to $O_1 \times O_2$, agreeing with the two given pieces on their intersections. Its C^m norm on every compact subset of the output product is at most $C_m M$. These constants are uniform when geometric margins, ellipticity, the required coefficient bounds, interpolation constants, and a positive lower bound for $\gamma_1 + \gamma_2 - 1$ are uniform. A specified output order requires only finitely many coefficient bounds.

Proof. Let $\mathcal{H}_1(D_1)$ be the closed subspace of $L^2(D_1; V_1)$ consisting of distributional L_1 -solutions. Restriction $R: \mathcal{H}_1(D_1) \rightarrow L^2(S_1; V_1)$ is compact by interior elliptic estimates and is injective by unique continuation. The spectral theorem for R^*R gives an orthonormal basis (w_j) of $\mathcal{H}_1(D_1)$ such that

$$(w_i, w_j)_{L^2(S_1)} = \mu_j^2 \delta_{ij}, \quad 0 < \mu_j \leq 1. \quad (3.2)$$

For every N there is a uniform constant C_N with

$$\mu_j \leq C_N (1 + j)^{-N}. \quad (3.3)$$

Indeed, multiply a solution by a cutoff equal to one near $\overline{S_1}$ and compactly supported in D_1 . Interior estimates bound its H^k norm by its $L^2(D_1)$ norm. Fourier truncation in a containing cube, component by component, gives an approximation of rank $O(A^n)$ and error $O(A^{-k})$. The minimax characterization of singular values yields $\mu_j \leq C_k j^{-k/n}$. The fixed fiber dimension changes only the constants.

For $y \in S_2$, expansion in the first variable gives

$$F(x, y) = \sum_j w_j(x) \otimes b_j(y), \quad \|b_j\|_{L^2(S_2; V_2)} \leq M.$$

Contracting the first tensor index against $\overline{w_j}$ extends the coefficients to D_2 :

$$b_j(y) = \mu_j^{-2} \int_{S_1} \sum_{\alpha} \overline{w_{j,\alpha}(x)} F_{\alpha}(x, y) dx, \quad \|b_j\|_{L^2(D_2; V_2)} \leq M \mu_j^{-1}. \quad (3.4)$$

Here $F_{\alpha}(x, y) \in V_2$ are the first-index components in an orthonormal basis of V_1 ; the displayed contraction leaves a vector in V_2 . Because the second system acts on that remaining index, $L_2 b_j = 0$. Restriction orthogonality proves that this formula agrees with the original coefficient on S_2 . The assumed interpolation gives

$$\|w_j\|_{L^2(O_1)} \leq C \mu_j^{\gamma_1}, \quad \|b_j\|_{L^2(O_2)} \leq C M \mu_j^{\gamma_2 - 1}.$$

Thus the tensor series converges absolutely in $L^2(O_1 \times O_2)$: the norm of its j th term is at most $C M \mu_j^{\gamma_1 + \gamma_2 - 1}$, and (3.3) makes these bounds summable. The limit solves both equations distributionally. Interior estimates for their elliptic sum give smoothness and the claimed bounds.

Agreement on the second arm requires a completeness check. Take $z \in L^2(S_1; V_1)$ orthogonal to the closure of $R\mathcal{H}_1(D_1)$. The V_2 -valued function

$$y \mapsto \int_{S_1} \sum_{\alpha} \overline{z_{\alpha}(x)} F_{\alpha}(x, y) dx$$

solves the second system and vanishes on S_2 , hence vanishes on connected D_2 . Every second-arm slice therefore belongs to the closed restricted range. Its expansion in the orthonormal basis w_j/μ_j is complete, with coefficients $\mu_j b_j(y)$. Consequently the series equals F on $S_1 \times O_2$; it converges there by the positive exponent γ_2 . On $O_1 \times S_2$ it is the original expansion, converging with exponent γ_1 . These establish both required agreements. Only upper bounds for μ_j have been used, so uniformity does not require continuous choices of singular bases or lower bounds on singular values. $\square$

The remaining issue is geometric: we must place two observed sets so that the interpolation exponents add to more than one. Strict Carleman weights will supply those exponents from a curvature condition on the boundary of the region where the product solution is known.

3.2 A geometric criterion for local extension

We next build the interpolation estimates needed in Lemma 3.1. For a metric g , call a smooth real function θ a *strict weight* at a point where $d\theta \neq 0$ if

$$\text{Hess}_g \theta(e, e) + \text{Hess}_g \theta(v, v) > 0 \quad \text{for all } v \perp e, |v|_g = 1, \quad e = \frac{\nabla^g \theta}{|\nabla^g \theta|_g}. \quad (3.5)$$

For the system $\mathbf{L} = -\Delta_g I + C^a \partial_a + D$, the strict Carleman estimate gives, after shrinking the coordinate neighborhood,

$$\tau^3 \|e^{\tau\theta} w\|_{L^2}^2 + \tau \|\nabla(e^{\tau\theta} w)\|_{L^2}^2 \leq C \|e^{\tau\theta} \mathbf{L} w\|_{L^2}^2, \quad \tau \geq \tau_0, \quad (3.6)$$

for compactly supported vector-valued w . All norms sum over the fiber components. The scalar form is the elliptic strict-weight estimate of the classical Carleman theory; see [11, Chapter XXVIII] and [13, Definition 8.3 and Theorem 9(a)]. We recall its energy proof to record the system dependence. First take one scalar component and $\mathbf{L} = -\Delta_g$, write $v = e^{\tau\theta} w$ and

$$e^{\tau\theta} \mathbf{L} e^{-\tau\theta} = A_\tau + B_\tau, \quad A_\tau = -\Delta_g - \tau^2 |d\theta|_g^2, \quad B_\tau = \tau(2\nabla^g \theta \cdot \nabla + \Delta_g \theta).$$

Here A_τ is self-adjoint and B_τ is skew-adjoint for metric volume. Integration by parts gives

$$\begin{aligned} \|(A_\tau + B_\tau)v\|^2 &= \|A_\tau v\|^2 + \|B_\tau v\|^2 + ([A_\tau, B_\tau]v, v), \\ ([A_\tau, B_\tau]v, v) &= 4\tau \int \text{Hess}_g \theta(\nabla v, \nabla \bar{v}) + 4\tau^3 \int \text{Hess}_g \theta(\nabla \theta, \nabla \theta) |v|^2 - \tau \int (\Delta_g^2 \theta) |v|^2. \end{aligned}$$

Choose a real constant m strictly between the negative of the least unit tangential Hessian value and the unit normal Hessian value. After shrinking the patch, these inequalities have a positive margin. Add $4m\tau(A_\tau v, v)$ to the commutator form. Its gradient coefficient becomes $4\tau(\text{Hess}_g \theta + mg)$, and its leading zero-order coefficient is

$$4\tau^3 |d\theta|_g^2 (\text{Hess}_g \theta(e, e) - m) > c\tau^3.$$

The tangential gradient form is positive. Its mixed and possibly negative normal terms cost at most $C\tau \|\nabla_e v\|^2$, and

$$\|\nabla_e v\|^2 \leq C\tau^{-2} \|B_\tau v\|^2 + C\|v\|^2.$$

Finally $|4m\tau(A_\tau v, v)| \leq \frac{1}{2} \|A_\tau v\|^2 + C\tau^2 \|v\|^2$. For large τ , the squared parts absorb the normal derivative cost and the positive zero-order term absorbs the remaining errors. This yields the scalar estimate. Summing it over components gives the estimate for $-\Delta_g I$. The conjugated matrix lower-order terms have norm at most $C(\|\nabla v\| + \tau\|v\|)$, so their squared norm is absorbed by the τ and τ^3 terms after increasing τ_0 . This proves (3.6) for the stated systems. The constants are uniform on compact sets of strict weights, with fixed positive strictness margins and bounded matrix coefficients. The scalar cutoff commutators used below act componentwise, and interior estimates for these systems have the same orders as in the scalar case.

The existence criterion below is the geometric criterion of [18, Section 3, “A geometric criterion for local extension”]. Its geometric proof depends only on the principal metrics. The system estimate just proved and Lemma 3.1 provide the analytic steps in the present setting. All gradients, lengths, orthogonality relations, and Hessians in its statement use the principal metrics g_1, g_2 and their product connection.

Proposition 3.2 (Product extension criterion). *Let ρ be smooth near $z_0 = (x_0, y_0)$, with $\rho(z_0) = 0$ and $d_x\rho, d_y\rho \neq 0$. Put*

$$a_i = \frac{\nabla_i \rho}{|\nabla_i \rho|^2}, \quad H = \text{Hess } \rho.$$

Suppose that at z_0

$$H((a_1, -a_2), (a_1, -a_2)) + H((v_1, v_2), (v_1, v_2)) > 0 \quad (v_i \perp a_i, |v_i| = |a_i|). \quad (3.7)$$

Every product solution given on $\{\rho > 0\}$ near z_0 extends to a neighborhood of z_0 , agreeing on a smaller part of that side. In fact, the construction uses only two compact product sets in $\{\rho > \varepsilon\}$ for some $\varepsilon > 0$. The neighborhoods and estimates on smaller neighborhoods are uniform under small smooth perturbations preserving the strict hypotheses, provided the data on those compact sets are bounded.

Proof. We first split the product Hessian condition into a strict weight in each factor. Their sum will lie below a defining function for the given side, with a quadratic gap. That gap places two intersecting product sets in the given side; the weights then yield interpolation exponents greater than one half, allowing Lemma 3.1 to apply.

Splitting the weight between the factors. We seek symmetric forms P_i satisfying

$$P_i(a_i, a_i) + P_i(v_i, v_i) > 0, \quad H + K d\rho^{\otimes 2} > P_1 \oplus P_2$$

for some $K > 0$, with the vectors v_i as in (3.7). After division by $|a_i|^2$, the first inequality is the strict-weight condition for a function with gradient $\nabla_i \rho$ and Hessian P_i . The second inequality will give the quadratic gap. To obtain these forms by convex separation, the dual tests are positive semidefinite forms D whose diagonal blocks are

$$d_i a_i^{\otimes 2} + S_i, \quad d_i \geq 0, \quad S_i \geq 0 \text{ on } a_i^\perp, \quad \text{tr } S_i = d_i |a_i|^2.$$

This follows by separating the open convex cone consisting of positive definite forms plus the allowed $P_1 \oplus P_2$; the individual dual cones are generated by $a_i^{\otimes 2} + v_i^{\otimes 2}$. Normalize $\text{tr } D = 1$. To obtain a suitable K , it suffices by compactness to prove $H : D > 0$ on tests with $D d\rho = 0$.

Represent such a D as a covariance matrix. The normal components measured by the two partial covectors are opposite. The diagonal-block condition makes each normal component uncorrelated with its own tangential component, hence with both tangential components. Their variances coincide, giving $d_1 = d_2 = d > 0$ and

$$D = d(a_1, -a_2)^{\otimes 2} + D_\perp, \quad \text{tr}(D_\perp)_{ii} = d|a_i|^2.$$

The case $d = 0$ would force $D = 0$. Among nonnegative tangential forms with these two fixed traces, every extreme point has rank one: on an image of rank $r \geq 2$, a nonzero symmetric perturbation preserves the two traces because $r(r+1)/2 > 2$, and small perturbations of both signs preserve positivity. The rank-one tests are precisely those in (3.7), multiplied by d . The asserted positivity follows. Compactness of the normalized tests then supplies K , as required.

Prescribe partial covectors $d_x\rho, d_y\rho$ and Hessians P_i for smooth functions θ_i vanishing at the respective points. In small coordinates centered there, Taylor expansion gives

$$\theta_1(x) + \theta_2(y) \leq \tilde{\rho}(x, y) - c(|x|^2 + |y|^2), \quad \tilde{\rho} = \rho + K\rho^2/2, \quad (3.8)$$

with $c > 0$. Each θ_i satisfies (3.5). The positive side of $\tilde{\rho}$ is the positive side of ρ in this neighborhood.

Placing the cross and obtaining interpolation. Use coordinates (s_i, z_i) with $s_i = \theta_i$. Fix a small $\kappa > 0$ so that $\phi(s_i) = s_i - \kappa s_i^2$ remains a strict weight. Choose successively a small lateral radius R , a height $l \ll cR^2$, and $e \ll \kappa l^2$. Take

$$D_i = \{-l < s_i < l + 4e, |z_i| < R\}.$$

Use a cutoff equal to one on the inner lateral half and for $-l + 3e < s_i < l + 2e$, supported in the cylinder and in $-l + 2e < s_i < l + 3e$. Let $S_i \Subset D_i$ include neighborhoods of its top and lateral derivative supports, where respectively

$$s_i > l + e/2 \quad \text{or} \quad |z_i| > R/3,$$

and an inner top slice near $s_i = l + 5e/4$. The two closed product sets $\overline{D_1 \times S_2}$ and $\overline{S_1 \times D_2}$ lie strictly in the given side. For a top slice, the sum of the two s coordinates is positive. For a lateral slice, it is greater than $-2l$, which is dominated by the quadratic gap in (3.8). Making the sets slightly smaller if necessary gives a common $\varepsilon > 0$ with $\rho > \varepsilon$ on their closures.

For an L_i -solution normalized by $\|w\|_{L^2(D_i)} = 1$, apply (3.6) to this cutoff times w . Interior estimates bound the bottom derivative terms using the large norm and the weight $\phi(-l + 3e)$. The remaining derivative terms use $\|w\|_{L^2(S_i)}$ and a weight at most $\phi(l + 3e)$. On a smaller inner cylinder beginning at $s_i = -e$, the weight is at least $\phi(-e)$. Balancing the two exponentials yields (3.1) with

$$\gamma_i = \frac{\phi(-e) - \phi(-l + 3e)}{\phi(l + 3e) - \phi(-l + 3e)} > \frac{1}{2}. \quad (3.9)$$

Indeed at $e = 0$ the quotient is $1/2 + \kappa l/2$. Values of the balancing parameter below τ_0 are covered by increasing the constant. The output cylinder may be chosen to include a connected tube from the origin to the indicated top slice.

Constructing and identifying the extension. Lemma 3.1 now gives a product solution on a neighborhood of z_0 . It agrees with the old solution there on the positive side: from a sufficiently near positive-side point, increase s_2 into the top slice. Since $\partial_{s_2}\rho > 0$ locally, the segment stays on that side and in the larger constructed cylinder. Agreement on the slice and unique continuation along a neighborhood of the segment prove agreement at the original point. All choices have strict margins. Keeping their finitely many compact sets and shrinking their output neighborhoods once proves the stated perturbation uniformity by (3.6), Lemma 3.1 and interior elliptic estimates. $\square$

3.3 Gluing along a compact family of surfaces

Proposition 3.2 provides a germ across one surface point. The following formulation records the geometry needed to combine these germs. In particular, agreement is proved on specified overlap components, rather than inferred from the existence of a cover.

Lemma 3.3 (Propagation through a compact cylinder). *Suppose a region in a product patch has smooth coordinates (z, a) near $B \times [a_-, a_+]$, where B is a compact base, possibly with boundary or corners. A single product solution is given on an open set $\mathcal{S}$ containing neighborhoods of the top and lateral boundary. Assume that $\mathcal{S}$ is upward closed in a at fixed z . If every level $a = \text{constant}$ satisfies Proposition 3.2 outside $\mathcal{S}$, with the positive side pointing toward increasing a , the solution extends to a neighborhood of the cylinder, agreeing on $\mathcal{S}$. For fixed coordinates, compact sets and strict reference inequalities, the extension has uniform compact-subset bounds under small smooth perturbations, in terms of bounds on finitely many compact subsets of the initially given region.*

Proof. Starting from the top, let a_* be the infimum of levels down to which an extension agreeing with the seed has been obtained. In the uniform version include uniform bounds in this property. At a non-seed point of $B \times \{a_*\}$, Proposition 3.2 uses compact data strictly above that level. Take a finite cover of the level by such output neighborhoods and seed neighborhoods. Their data are contained above $a_* + \eta$ for one $\eta > 0$, so an already reached level supplies them with the needed bounds.

Shrink the outputs in (z, a) coordinates to boxes $B_i \times (a_* - d_i, a_* + d_i)$. Every connected component of an overlap contains vertical segments to a common height greater than a_* . Both germs equal the preceding extension there. Unique continuation for $\mathbb{L}_1^x + \mathbb{L}_2^y$ makes them identical on that overlap component. The same argument gives agreement with $\mathcal{S}$, because a vertical segment moving upward from a seed point stays in the seed. Boundary neighborhoods use the given solution. A finite subcover has a positive minimum output width and therefore passes the putative last level, or includes the endpoint a_- . This proves continuation on the whole cylinder. The same finite constructions retain all strict margins under small perturbations. Their Carleman, series and interior estimates prove the uniform assertion. Data bounded up to a limiting surface are never required: each local construction uses compact sets strictly above it. $\square$

3.4 Initialization at a fixed distance from the diagonal

We now return to the connection Laplacians. In common coordinate patches centered at 0, let (g_j, A_j) be smooth metrics and unitary connection matrices obtained from interior patches of compact Dirichlet manifolds. Use fixed local unitary frames, and let G_j denote their matrix Dirichlet Green kernels, normalized with metric volume. Suppose that both coefficient pairs agree on an open set W containing the lower side of a smooth hypersurface through 0, and that $G_1(x, y) = G_2(x, y)$ for distinct $x, y \in W$. After a common linear coordinate change, assume

$$g_1(0) = g_2(0) = I_n, \quad \{x_n < q(x')\} \subset W \text{ near } 0, \quad q(0) = 0, \quad dq(0) = 0,$$

for a smooth function q . No regularity of the rest of ∂W is assumed. For a small spatial dilation parameter $\lambda > 0$, put

$$\begin{aligned} g_{j,\lambda}(z) &= g_j(\lambda z), & A_{j,\lambda}(z) &= \lambda A_j(\lambda z), & W_\lambda &= \lambda^{-1}W, \\ G_{j,\lambda}(x, y) &= \lambda^{n-2}G_j(\lambda x, \lambda y). \end{aligned} \tag{3.10}$$

On each fixed bounded patch the metrics tend smoothly to the Euclidean metric. The corresponding dilated systems have uniformly bounded smooth lower-order coefficients on every fixed bounded set, and W_λ contains every fixed compact subset of $\{z_n < 0\}$ for sufficiently small λ . Initially define, off the diagonal,

$$\mathcal{G}_\lambda(x, y) = \begin{cases} G_{1,\lambda}(x, y), & y \in W_\lambda, \\ G_{2,\lambda}(x, y), & x \in W_\lambda. \end{cases} \tag{3.11}$$

The pieces agree on their overlap. They solve the first connection Laplacian in x and the conjugate second connection Laplacian on the second index in y , in the convention at the beginning of the section. Indeed, on the observed factor its two coefficient systems agree. Thus (3.11) is a product solution of the systems already treated.

Proposition 3.4 (Fixed-separation initialization). *In this setting, let $n \geq 3$ and fix $B > 0$, $d > 0$ and $\eta > 0$. For all sufficiently small λ , the data (3.11) have a product-solution continuation to a neighborhood of*

$$\{(x, y) : |x|, |y| \leq B, \ |x - y| \geq d\}.$$

Every fixed C^m norm on a smaller neighborhood is bounded uniformly in λ . It agrees with $G_{1,\lambda}$ when $y \in W_\lambda$ and with $G_{2,\lambda}$ when $x \in W_\lambda$, on the corresponding overlaps. In particular these agreements hold on the fixed truncated lower strips $y_n < -\eta$ and $x_n < -\eta$, respectively. The smallness threshold for λ and the constants may depend on B, d, η, m .

Proof. For the final comparison with the full original cross, fix at the outset

$$\hat{B} = B + \eta + 4, \quad \hat{d} = \min(d/2, 1/4).$$

We construct the continuation for these enlarged location bounds and smaller separations, and then restrict to the stated target. We first cross the limiting plane, then deform variable-radius tubes to reach those separations. This follows the two-stage construction of [18, Section 3, “Fixed-separation initialization”]. We first choose the Euclidean geometric ratios, including the opening of the region reached in the first stage. We next choose the entire tube cylinder and a bounded positive-separation range containing it. The first-stage constructions are then made on that range. Only after all these fixed sets have been chosen do we decrease λ .

Crossing the plane and obtaining a common seed. We first extend slightly across the plane in one variable while keeping the other at a comparable positive separation. The first interpolation exponent can be made close to one; this compensates for the fixed positive exponent obtained by propagation in the separated second variable.

For the Euclidean equations, the following choices are invariant under translations parallel to the plane and under a common change of spatial scale. We make the geometric choices on a unit-scale template. They therefore give an opening constant $c_0 > 0$ independent of the bounded range on which the construction is subsequently used. On any fixed compact range of centers and scales $0 < s_{\min} \leq s \leq s_{\max}$, the same strict weights and geometric fractions work for the dilated systems once λ is sufficiently small. The lower-order coefficients are bounded uniformly there (and converge to zero under the connection dilation).

Fix a point x_0 of height zero and a separation scale $s > 0$. In the second factor take

$$D_2 = \{s/2 < |y - x_0| < 2s\}$$

and an observed ball S_2 about $x_0 - se_n$. In the first take a ball of radius $3\vartheta_0 s$ centered at $x_0 - \vartheta_0 se_n$, with $\vartheta_0 > 0$ fixed small enough that the two large domains are separated. Its observed ball has radius $(1 - \alpha)\vartheta_0 s$, with $\alpha > 0$ to be chosen. These observed balls lie strictly in the lower half-space. On a fixed compact scale range their heights are bounded above by a common negative number, since $s \geq s_{\min} > 0$. Thus, after a finite covering, one choice of small λ places every observed ball in W_λ with room. No neighborhood of the whole limiting plane is assumed to belong to W_λ .

There is a two-constants estimate from S_2 to a slightly enlarged closed annulus $4s/5 \leq |y - x_0| \leq 6s/5$, with an exponent $\gamma_2 > 0$ independent of α . Here is a direct way to obtain it with room for perturbations. On a Euclidean annulus the weight $|z - z_*|^{-1}$ is strict: its radial Hessian is $2|z - z_*|^{-3}$ and each tangential Hessian value is $-|z - z_*|^{-3}$. Radial cutoffs in (3.6), followed by exponential balancing, propagate smallness from an inner ball to an intermediate ball using the norm on an outer ball. A finite chain of overlapping balls with enlarged balls inside D_2 connects S_2 to the target annulus. The annulus is connected since $n \geq 3$. Multiplying the finitely many positive exponents gives the asserted γ_2 . The same weights and margins work for sufficiently small perturbations of the Euclidean metric.

In the first factor use this radial argument centered at $x_0 - \vartheta_0 se_n$. Keep the outer cutoff a fixed distance beyond radius $\vartheta_0 s$, and put the inner cutoff just inside radius $(1 - \alpha)\vartheta_0 s$.

Take a concentric target ball of radius $(\vartheta_0 + c)s$, with $c > 0$ small; it contains the observed ball and a ball of radius cs about x_0 . To see the resulting exponent, let R_{in} be an inner cutoff radius, chosen just inside the observed radius, let $R_{\text{tar}} = (\vartheta_0 + c)s$, and let $R_{\text{out}} > R_{\text{tar}}$ be a fixed starting radius for the outer derivative support. With $\theta(R) = R^{-1}$, exponential balancing gives

$$\gamma_1 = \frac{\theta(R_{\text{tar}}) - \theta(R_{\text{out}})}{\theta(R_{\text{in}}) - \theta(R_{\text{out}})}.$$

The inner core is bounded directly by the observed norm. As α and c decrease, choose the inner cutoff so that both R_{in} and R_{tar} approach $\vartheta_0 s$, while R_{out} stays a fixed distance beyond that radius. Thus $\gamma_1 \rightarrow 1$. Choose these constants so that $\gamma_1 + \gamma_2 > 1$. Lemma 3.1 extends the kernel to a small ball about x_0 times the target annulus. On these fixed separated sets the rescaled Green kernels and their derivatives are uniformly bounded: the same bounds hold entrywise for the smooth systems used here. To see this, an interior inverse parametrix has order -2 and frequency pieces bounded by $C_N 2^{j(n-2)}(1 + 2^j|x - y|)^{-N}$, up to a smooth kernel. Summing gives the local bound $C|x - y|^{2-n}$ for $n \geq 3$. The normalization in (3.10) preserves this bound; interior estimates for the uniformly elliptic dilated equations give all fixed derivative bounds away from the pole.

The extensions agree with both lower pieces on their actual comparison domains. For $y \in S_2$, the series equals $G_{1,\lambda}$ for all x in its first output ball. Fix any such x in a truncated lower half-space contained in W_λ . On S_2 both variables then lie in W_λ , so this value also equals $G_{2,\lambda}$. Unique continuation for the second system on the connected output annulus proves agreement with $G_{2,\lambda}$ throughout that annulus. In the other direction, fix y in the output annulus and in W_λ . On the first observed ball S_1 , agreement with the second arm gives $G_{2,\lambda} = G_{1,\lambda}$. Unique continuation for the first system on its connected output ball gives agreement with $G_{1,\lambda}$ there. Each comparison stays a positive distance from the diagonal.

We make the overlap comparison precise because the two variables may play different roles in two local constructions. Inside the output just obtained choose a larger comparison product of the form

$$P(x_0, s) = B(x_0, \kappa s) \times \{(1 - \zeta)s < |y - x_0| < (1 + \zeta)s\},$$

where $\kappa, \zeta > 0$ are fixed geometric fractions. The first ball is contained in the actual concentric first output ball, and the annulus is contained in the actual second output. Keep the full original outputs for the preceding lower-piece comparisons. For coverage retain only

$$P'(x_0, s) = B(x_0, \epsilon_1 s) \times \{(1 - \zeta')s < |y - x_0| < (1 + \zeta')s\}, \quad 0 < \epsilon_1 \ll \kappa, \quad 0 < \zeta' < \zeta.$$

The transposed products are used when y is the near-plane variable. If a point belongs to two retained products, their scales are comparable to each other and to $|x - y|$, with fixed constants. Choose ϵ_1 small enough, relative to κ and $\zeta - \zeta'$, that a downward translation of size at most a fixed multiple of ϵ_1 times either scale stays inside each larger comparison product whenever it acts on a retained near-plane variable. This follows directly from the triangle inequality for the ball and annular distances. On the chosen compact scale range take $\eta_0 > 0$ much smaller than the remaining margins times s_{min} , and with $\eta_0 \leq \eta$.

For products of the same orientation, lower only their common near-plane variable by

$$h = \max(x_n, 0) + 2\eta_0$$

(or the analogous expression in y). The other variable remains fixed. For products of opposite orientations, both x and y lie in retained near-plane balls; lower both by

$$h = \max(x_n, y_n, 0) + 2\eta_0.$$

The scale comparability and the preceding choices keep the entire path in the intersection of the two larger comparison products. Its endpoint has the near-plane variable below $-\eta_0$ in the first case, and both variables below $-\eta_0$ in the second. For small enough λ these endpoint neighborhoods lie in the original lower cross, where the preceding comparisons identify both germs. Unique continuation for the elliptic sum on the component of the larger intersection containing the path identifies the germs at the starting point. The path need not remain in the retained products. This proves agreement on every retained overlap without a connectedness assumption on that overlap.

Apply the same construction with the variable roles and their tensor factors interchanged, and then return the values to the original tensor order. The resulting germs solve the same two equations as the first orientation. The overlap comparison therefore applies to them and produces, for some fixed $c_0 > 0$, a single-valued continuation in

$$\min(x_n, y_n) < c_0|x - y|, \quad (3.12)$$

on every fixed compact range of bounded locations and positive separations. The estimates are uniform in λ on that range. For coverage near the plane, take $x_0 = (x', 0)$ and $s = |y - x_0|$ when the first variable has the smaller nonnegative height; then x is in the retained first ball and y in the retained annulus if c_0 is small; for instance $c_0/(1 - c_0) < \epsilon_1$ guarantees the required ball inclusion. Points already below the plane are covered by the same construction near it or by the given cross farther below it. The preceding overlap argument makes these choices compatible.

Sweeping tubes inward from the common seed. The region (3.12) is now the seed for the second stage. At fixed center y and direction ω , we move x along the ray from y : large radii lie in the seed, and decreasing the radius will reach the target pairs. Choose $M > \widehat{B} + 3$ and introduce

$$t = M + y_n + e_0|y'|^2, \quad r_a(y) = at^{1+\sigma}, \quad x = y + r_a(y)\omega, \quad |\omega| = 1, \quad (3.13)$$

where $\sigma, e_0 > 0$ will be small. The base is

$$y_n \geq -\widehat{B} - 1, \quad t \leq T_0, \quad \omega \in \mathbb{S}^{n-1},$$

and a decreases through a fixed interval $[a_{\min}, a_{\max}]$. The term $e_0|y'|^2$ makes this base compact. The parameter a decreases while (y, ω) remains fixed, as required by the compact-cylinder propagation lemma. Put $m = M - \widehat{B} - 1 > 2$. Reserve the bounds $0 < \sigma \leq 1$, $e_0 \leq (1 + \widehat{B}^2)^{-1}$, and $e_0 T_0 \leq 1/4$. Once σ is chosen below, take

$$0 < a_{\min} < \frac{\widehat{d}}{2(M + \widehat{B} + 1)^{1+\sigma}}, \quad a_{\max} > \max\left(a_{\min}, \frac{2}{c_0 m^\sigma}\right),$$

and then choose

$$T_0 > 2(M + \widehat{B} + 1), \quad c_0 a_{\min} T_0^\sigma > 2.$$

For every target center, $t \leq M + \widehat{B} + 1$; hence its target separations exceed $r_{a_{\min}}(y)$. On the initial level, $c_0 r_{a_{\max}} > 2t > y_n$, so this level lies in (3.12). The bottom base boundary has $y_n < 0$, and on the face $t = T_0$ we have $c_0 r_a > 2t > y_n$ for all allowed a . Thus the initial level and both base boundary faces are supplied by the seed. The power $1 + \sigma > 1$ is what permits this last inequality at the top face. This known region is upward closed in a : since $\min(x_n, y_n) = y_n + r \min(\omega_n, 0)$, its defining inequality is $y_n < (c_0 - \min(\omega_n, 0))r$, whose radius coefficient is positive.

It remains to check the strict extension criterion outside that region. At the Euclidean metrics use

$$\rho = \frac{1}{2}(|x - y|^2 - r_a(y)^2), \quad b = \nabla r_a.$$

Away from $\omega + b = 0$, multiply the vectors in (3.7) by the common positive factor r_a and write their real and imaginary parts as

$$\begin{aligned} U_1 &= \omega + iv, & |v| &= 1, \quad v \perp \omega, \\ U_2 &= \frac{\omega + b}{|\omega + b|^2} + iw, \quad w \perp (\omega + b), & |w| &= |\omega + b|^{-1}. \end{aligned} \quad (3.14)$$

The identity $\omega \cdot (U_1 - U_2) = b \cdot U_2$ cancels the normal part of the distance Hessian. Consequently the test is

$$|\pi_{\omega^\perp}(U_1 - U_2)|^2 - (r_a \partial^2 r_a)(U_2, \overline{U_2}). \quad (3.15)$$

For bounded b , outside fixed small neighborhoods of $b = -\omega$ and $b = -2\omega$, the first term is at least $c|b|^2$. To see this, a zero forces the real part of U_2 to be parallel to ω , and equality of the projected imaginary lengths gives $|\omega + b| = 1$. Thus $b = 0$ or $b = -2\omega$. Near $b = 0$, the real projected difference controls $|\pi_{\omega^\perp} b|$ to first order; also

$$|\pi_{\omega^\perp} w| = 1 - \omega \cdot b + O(|b|^2),$$

so the imaginary projected difference controls $|\omega \cdot b|$ to first order. This proves the quadratic lower bound near zero; compactness proves it on the remaining range.

Outside (3.12), $y_n \geq c_0 r_a$, so $r_a/t \leq 1/c_0$ and

$$b = (1 + \sigma)(r_a/t)\nabla t$$

has $|b| \leq 4/c_0$ under the reserved bounds. This fixes a compact range for b before σ , the a -interval, T_0 , and e_0 receive their final values. When ∇t is close to e_n , the two excluded neighborhoods already lie in (3.12). Indeed,

$$\frac{x_n}{r_a} \leq \frac{(1 + \sigma)|\nabla t|}{|b|} + \omega_n,$$

which is arbitrarily small near $b = -\omega$ and negative near $b = -2\omega$ as σ and the gradient error tend to zero. On the complement $|U_2|$ is bounded and differentiation of (3.13) gives

$$r_a |\partial^2 r_a| \leq C(\sigma + T_0 e_0) |b|^2. \quad (3.16)$$

Here the choices can be made in a definite order. First choose a neighborhood radius $\delta_0 > 0$ for $b = -\omega$ and $b = -2\omega$, in terms of c_0 , and upper bounds on σ and $|\nabla t - e_n|$ so that their closures satisfy $x_n/r_a < c_0/2$. On the complementary bounded b -range let $c > 0$ be the constant in the quadratic lower bound. There $|U_2|^2 \leq 2/\delta_0^2$. Direct differentiation sharpens (3.16) to

$$\frac{r_a |\partial^2 r_a|}{|b|^2} \leq \sigma + 2e_0 T_0.$$

Choose $\sigma > 0$ small enough that its contribution to this bound is less than $c\delta_0^2/8$. Choose the a -interval and T_0 as above, and finally take $e_0 > 0$ sufficiently small that all reserved bounds hold, $|\nabla t - e_n| \leq 2\sqrt{e_0 T_0}$ has the required smallness, and $\sigma + 2e_0 T_0 < c\delta_0^2/4$. The second term of (3.15) is then at most $(c/2)|b|^2$ in absolute value. The whole test is bounded below by

$$\frac{c}{2}|b|^2 \geq \frac{c}{2}(1 + \sigma)^2 a_{\min}^2 m^{2\sigma} > 0$$

where continuation is needed. Also the two partial gradients of ρ have lengths at least r_a and $\delta_0 r_a$, respectively. This supplies fixed strict margins before any metric perturbation.

All these sets are bounded and stay at separation at least $a_{\min}(M - \widehat{B} - 1)^{1+\sigma} > 0$. Smooth convergence of the metrics therefore preserves the strict tests for sufficiently small λ . At this point the tube parameters and its compact range have all been fixed. Apply the first-stage construction on a slightly larger bounded range of locations and separations containing this cylinder, and then choose λ small enough for both its data and the strict tube tests. Lemma 3.3 continues the kernel through the cylinder, with uniform estimates and agreement on the seed. A target pair has $a = |x - y|/t^{1+\sigma} \geq a_{\min}$; if $a > a_{\max}$ it is already in the coarse region. Hence every target pair is covered.

It remains to identify this extension with both pieces of the full cross (3.11), including parts of W_λ not in a truncated half-space. For $|y| \leq B$ put

$$D_y = B(0, \widehat{B}) \setminus \overline{B(y, \widehat{d})}, \quad A = B(-(B + \eta + 2)e_n, 1/4).$$

The excluded closed ball lies strictly inside $B(0, \widehat{B})$, so D_y is connected; it contains the fixed lower ball A , which is disjoint from every excluded ball and lies below $x_n = -\eta$. For fixed $y \in W_\lambda$ in the target range, the constructed kernel and $G_{1,\lambda}$ solve the same first-variable system on D_y . On A the constructed kernel equals $G_{2,\lambda}$, which equals $G_{1,\lambda}$ because both variables are in W_λ . Unique continuation on D_y proves agreement on the target overlap. Interchanging the variables proves agreement with the other piece. The inequalities defining $\widehat{B}$ and $\widehat{d}$ leave open room around the original target, so this also gives the claimed neighborhood and its compact-subset estimates. $\square$

The constants in Proposition 3.4 may deteriorate as $d \downarrow 0$. Its role is to initialize the sphere construction at fixed positive separations. Lemma 3.3 will subsequently give existence at each positive radius of a shrinking family. The separate estimate on that family's transfer density, rather than an estimate obtained by iterating this continuation, will control the shrinking limit.

4 Shrinking spheres and harmonic transfer

We now work with the contact data of Proposition 2.5. The two systems agree on the lower side of a smooth hypersurface through the origin, and their Green matrices agree when both arguments are on that side. Our aim is to represent a map between their local harmonic sections by integration over small spheres. Product continuation constructs this map at each positive radius. A separate estimate in the next section will control its density as the radii collapse.

The sphere geometry and radius profiles below are adapted from [18, Section 4]. We give their proofs, including the strict Hessian calculation, and formulate the resulting transfer for the present matrix equations. The coordinates need not be harmonic.

4.1 A concave depth and the radius profiles

Apply the spatial dilation (2.10). In this section we suppress the dilation subscript on g_j, A_j, G_j, F_j and on the paired functions, while retaining λ for the dilation parameter. On every fixed bounded coordinate region, $g_j \rightarrow I$ smoothly and $A_j \rightarrow 0$ smoothly as $\lambda \downarrow 0$. The harmonic frames and their inverses have uniform smooth bounds there. All bounded regions and positive separation scales used below are fixed before the final upper bound on λ is chosen.

Put

$$Y = \{y = (y', y_n) : |y'| \leq 1, -1 \leq y_n \leq 1\}, \quad t_0(y) = y_n + |y'|^2.$$

For a sufficiently large fixed L , define

$$\gamma = e^{-2Lt_0} g_1, \quad a = \gamma^{-1}, \quad Q = \frac{ng_2^{-1}}{\text{tr}(\gamma g_2^{-1})}, \quad h = Q - a. \quad (4.1)$$

These tensors are defined on a fixed neighborhood of Y , also used for the first-variable points. Specify the scalar multipliers in the harmonic-frame convention (2.11) by

$$\mathcal{L}_j v = -c_j F_j^{-1} L_j (F_j v), \quad c_1 = e^{2Lt_0}, \quad c_2 = \frac{n}{\text{tr}(\gamma g_2^{-1})} > 0. \quad (4.2)$$

Their second-derivative coefficient matrices are a and Q , respectively, and their zeroth-order terms vanish because the columns of F_j are harmonic. These changes leave the solution spaces unchanged after the frame identification. For product continuation we use the positive-principal-symbol convention: $-\mathcal{L}_j$ in harmonic frames, or equivalently $c_j L_j$ in unitary frames. The corresponding principal metrics are γ and Q^{-1} . The normalization ensures

$$\text{tr}(\gamma h) = 0, \quad h \longrightarrow 0 \quad \text{in every fixed smooth norm as } \lambda \downarrow 0. \quad (4.3)$$

Define a depth function by

$$t(y) = \int_{1/8}^{t_0(y)} e^{-2Ls} ds. \quad (4.4)$$

It is negative near the origin and has nonvanishing differential. Its useful feature is strict concavity for the reference metric:

$$\text{Hess}_\gamma t = e^{-2Lt_0} (\text{Hess}_{g_1} t_0 - L |dt_0|_{g_1}^2 g_1) \leq -c\gamma. \quad (4.5)$$

Indeed the conformal connection formula cancels the two $dt_0 \otimes dt_0$ terms. The norm of dt_0 is bounded below on the fixed region, whereas $\text{Hess}_{g_1} t_0$ is uniformly bounded for small λ . Choosing L first proves the last inequality. All constants describing this reference geometry are henceforth fixed.

For an integer $p_* \geq 2$, an amplitude $R_* > 0$, and $0 < \delta \leq 1$, put

$$\ell_\delta(t) = \delta \log(1 + e^{t/\delta}), \quad r_\delta(y) = R_* \ell_\delta(t(y))^{p_*}, \quad f_* = \log r_\delta, \quad \beta = |df_*|_\gamma. \quad (4.6)$$

We use the γ -geodesic ball with center y and radius $r_\delta(y)$. These balls shrink to a point on $\{t \leq 0\}$, including a neighborhood of the contact point. At positive depths bounded away from zero, their radii retain a positive lower bound. This will let us place cutoff derivatives where the transfer is already controlled.

Lemma 4.1 (Uniform profile bounds). *For the geometry just fixed and all sufficiently small λ , the profiles (4.6) satisfy*

$$\beta \geq cp_*, \quad \text{Hess}_\gamma f_* \leq -c\beta\gamma, \quad |\partial^j f_*| \leq C_j \beta^j \quad (j \geq 1), \quad (4.7)$$

uniformly in $0 < \delta \leq 1$. The constants c, C_j are independent of $p_ \geq 2$. For fixed p_*, R_* , the functions r_δ^{1/p_*} have a uniform Lipschitz bound. The radii increase strictly with δ , converge to zero where $t \leq 0$, and obey*

$$\sup_{0 < \delta \leq 1, y \in Y} r_\delta \beta^2 \longrightarrow 0 \quad \text{as } R_* \downarrow 0 \quad (4.8)$$

for each fixed p_ .*

Proof. Write $a_\delta = (\log \ell_\delta)'$. With $s = t/\delta$,

$$a_\delta(t) = \delta^{-1} A(s), \quad A(s) = \frac{e^s}{(1 + e^s) \log(1 + e^s)}.$$

The function A is positive and decreasing: setting $u = e^s$ gives

$$A'(s) = \frac{u(\log(1 + u) - u)}{(1 + u)^2 \log(1 + u)^2} < 0.$$

At the negative end A tends to one, and at the positive end it is comparable to $(1 + s)^{-1}$. Differentiating its convergent expressions at the two ends gives $|A^{(j)}| \leq C_j A^{j+1}$; positivity gives the same bounds on the remaining compact interval. Consequently, on the fixed range of t ,

$$a_\delta \geq c > 0, \quad a'_\delta \leq 0, \quad |\partial_t^j \log \ell_\delta| \leq C_j a_\delta^j, \quad a_\delta \leq \ell_\delta^{-1}.$$

Since $|dt|_\gamma$ is bounded above and below, $\beta = p_* a_\delta |dt|_\gamma$. The identity

$$\text{Hess}_\gamma f_* = p_* a_\delta \text{Hess}_\gamma t + p_* a'_\delta dt \otimes dt$$

and (4.5) prove the first two bounds in (4.7). The chain rule gives the derivative bounds; their constants can be independent of p_* because $p_* a_\delta^j \leq p_*^j a_\delta^j$ for $j \geq 1$.

The estimate $0 < \ell'_\delta < 1$ proves the Lipschitz assertion. Moreover,

$$\partial_\delta \ell_\delta = \log(1 + e^s) - \frac{se^s}{1 + e^s} > 0.$$

The displayed expression tends to zero at both ends and its derivative in s is $-se^s/(1 + e^s)^2$, so it is strictly positive at finite s . The limit of ℓ_δ is $\max(t, 0)$. Finally,

$$r_\delta \beta^2 \leq C R_* p_*^2 \ell_\delta^{p_*-2},$$

and ℓ_δ is uniformly bounded on the fixed depth interval. This proves (4.8). $\square$

4.2 The core, cutoff, and order of choices

Choose $t_b > 0$ so small that $3t_b < t(1/4)$, where $t(1/4)$ denotes the value of the increasing function in (4.4) at $t_0 = 1/4$. Define

$$\begin{aligned} K &= \{y \in Y : y_n \geq -1/2, t(y) \leq 2t_b\}, \\ \chi &\in C_c^\infty(Y^\circ), \quad \chi = 1 \text{ near } K, \\ \text{supp}(d\chi) &\subset \{y_n < -1/4\} \cup \{t > t_b\}. \end{aligned} \tag{4.9}$$

Such a cutoff exists. Indeed, on K one has $t_0 < 1/4$, and thus $|y'|^2 < 3/4$ and $y_n < 1/4$; hence $K \Subset Y^\circ$. Multiply a cutoff changing from zero to one between $y_n = -3/4$ and $y_n = -1/2$ by a cutoff of t changing from one to zero between $2t_b$ and $3t_b$, moving its endpoints slightly to leave room around K . The inequality $3t_b < t(1/4)$ keeps the support away from the lateral and upper faces of Y . On the second cutoff region,

$$r_\delta \geq R_* t_b^{p_*} > 0. \tag{4.10}$$

Every point of K can be joined to the known lower region by the vertical segment with the same y' and initial height $-1/2$. That segment stays in K ; both t and r_δ are nondecreasing along it. For sufficiently small λ , its initial point and a neighborhood of it lie in the known side.

We will use the following temporary metric comparison:

$$H_8(y) := \sum_{|\alpha| \leq 8} |\partial^\alpha h(y)| \leq \varepsilon r_\delta(y) \quad (y \in Y). \quad (4.11)$$

Coordinate norms in this definition are uniformly equivalent to the reference-metric norms. This condition will later be improved and then removed by a continuity argument.

Remark 4.2 (Parameter choices). The reference geometry, t_b , K , and χ are fixed first. Choose p_* sufficiently large and then $\varepsilon > 0$ sufficiently small. The lower bounds on p_* and upper bounds on ε used below depend only on the fixed geometry and coefficient bounds, before R_* and the final dilation are chosen. Choose R_* so that every ball lies in a normal neighborhood with fixed coordinate room and

$$r_\eta |d \log r_\eta|_\gamma^2 \leq \varepsilon \quad (0 < \eta \leq 1, y \in Y). \quad (4.12)$$

The fixed-separation scales determined by these choices are positive, although they may be small. Only then reduce λ to meet the fixed-separation initialization and the coefficient bounds. The parameter δ remains free subject to (4.11). In particular the upper bound on λ does not depend on δ .

4.3 The strict Hessian test for a moving sphere

The difficulty in approaching the diagonal is that the radius varies with the center. The leading terms from its gradient cancel in the product Hessian test. The strict concavity of its logarithm provides the positive remainder.

Lemma 4.3 (Strict sphere geometry). *Let $r = r_\delta$ satisfy (4.7), (4.11), and (4.12). For p_* sufficiently large and ε sufficiently small, the hypersurface*

$$\rho(x, y) = \frac{1}{2}(d_\gamma(x, y)^2 - r(y)^2) = 0$$

satisfies the strict criterion of Proposition 3.2 for principal metrics γ in the first factor and Q^{-1} in the second. Both partial gradients are nonzero, and the positive side is the exterior of the moving ball.

Proof. Put $b = \nabla^\gamma r$, so that $|b| = r\beta$. Parallel transport along the radial γ -geodesic identifies the endpoint tangent spaces; let ω be its unit direction from y to x . Multiply the normalized vectors in Proposition 3.2 by the common factor r . Their real and imaginary parts can be written as

$$U_1 = \omega + iv, \quad v \perp \omega, \quad |v| = 1, \quad U_2 = q + iw.$$

All lengths in the following calculation use γ . Set $s = \omega + b$. In a γ -orthonormal frame at y , the exact conditions on the second vector are

$$q = \frac{Qs}{s^T Qs}, \quad s \cdot w = 0, \quad w^T Q^{-1} w = \frac{1}{s^T Qs}. \quad (4.13)$$

Here the matrix of a is the identity. Therefore

$$s \cdot U_2 = 1, \quad q = \frac{s}{|s|^2} + O(|h|), \quad |w| = |s|^{-1} + O(|h|). \quad (4.14)$$

The assumptions make b and h small. Both partial gradients of ρ are consequently nonzero, and $|U_2|$ is bounded above and below.

With the product γ -connection, the endpoint Hessian of half the squared distance has value

$$|U_1 - U_2|^2 + O(r^2)(|U_1|^2 + |U_2|^2). \quad (4.15)$$

To obtain the error, parametrize the radial geodesic on $[0, 1]$. In a parallel frame its Jacobi field with prescribed endpoint values differs from the affine interpolant by $O(r^2)$ in C^1 , because the curvature is bounded and the velocity has size r . The second variation formula gives (4.15). Replacing the second connection by that of Q^{-1} contributes only $O(r^2)$: its difference from the γ -connection is $O(|h| + |\partial h|) = O(\varepsilon r)$, while $|d\rho| = O(r)$.

The relation $s \cdot U_2 = 1$ implies $\omega \cdot (U_1 - U_2) = b \cdot U_2$. Also

$$\text{Hess}_\gamma(r^2/2) = 2 dr \otimes dr + r^2 \text{Hess}_\gamma f_*.$$

Thus the Hessian sum in the criterion, multiplied by r^2 , equals

$$|\pi_{\omega^\perp}(U_1 - U_2)|^2 - |b \cdot U_2|^2 - r^2 \text{Hess}_\gamma f_*(U_2, \overline{U_2}) + O(r^2). \quad (4.16)$$

For a real bilinear form, evaluation on a complex vector and its conjugate means the sum of the evaluations on its real and imaginary parts, as in the criterion.

We record the cancellation quantitatively. The real projected difference is $-\pi_{\omega^\perp} b + O(|b|^2 + |h|)$. The imaginary part obeys

$$|\pi_{\omega^\perp} w| = 1 - \omega \cdot b + O(|b|^2 + |h|).$$

Comparing this length with $|v| = 1$ and squaring gives

$$|\pi_{\omega^\perp}(U_1 - U_2)|^2 \geq |b|^2 - C(|b|^3 + |b||h| + |h|^2).$$

The real and imaginary directions of U_2 are almost orthonormal; hence

$$|b \cdot U_2|^2 \leq (1 + C(|b| + |h|))|b|^2.$$

Using $|b| = r\beta$, $|h| \leq \varepsilon r$, and (4.7), expression (4.16) is bounded below by

$$cr^2\beta - C(r^2 + r^3\beta^3 + \varepsilon r^2\beta + \varepsilon^2 r^2). \quad (4.17)$$

Divide the error terms by $r^2\beta$. Their ratios are bounded by C/β , $Cr\beta^2$, $C\varepsilon$, and $C\varepsilon^2/\beta$, respectively. Increasing p_* controls the first, and then decreasing ε controls the others by (4.12). This proves strict positivity. $\square$

Proposition 4.4 (Continuation to the sphere family). *Make the choices of Remark 4.2. For all sufficiently small λ and every $0 < \delta \leq 1$ satisfying (4.11), the mixed Green matrix extends to a neighborhood of*

$$\{(x, y) : y \in Y, d_\gamma(x, y) = r_\delta(y)\}.$$

It solves the first system in x and the conjugate second system in the second tensor index. On the lower center region $y_n < -1/4$ it agrees with G_1 , and on the prescribed positive-depth region $t > t_b$ it agrees with the fixed-separation continuation of Proposition 3.4. The extension is smooth for each positive δ .

Proof. Deform η from 1 down to δ , using (y, ω) in the unit sphere bundle of γ as base coordinates and

$$x = \exp_y^\gamma(r_\eta(y)\omega).$$

The base over the closed cylinder Y is compact. The radius has strictly positive derivative in η , so these variables give cylinder coordinates on every fixed interval $\delta \leq \eta \leq 1$. The positive side of a level is the direction of increasing η . Since $r_\eta \geq r_\delta$, condition (4.11) holds at every intermediate radius. Lemma 4.3 applies there.

At $\eta = 1$ the separations have a positive minimum on Y . The same is true throughout $t > t_b$ by (4.10). Proposition 3.4 therefore supplies the top and positive-depth data, with open room and uniform bounds at any fixed order. In the lower strip $y_n < -1/4$, use the original kernel G_1 ; this strip lies in the dilated known side for small λ . The initialization agrees with this kernel on their overlap.

These two base regions cover every lateral face of Y : its bottom face is in the lower strip, and on a remaining lateral or upper face outside that strip one has $t_0 \geq 3/4$. Both regions are independent of η and hence upward closed. Together with a neighborhood of the top level they supply exactly the seed required by Lemma 3.3. That lemma and the strict sphere test extend the kernel through the whole cylinder, preserving agreement on the seed. Normal-neighborhood room and positive radii on each fixed interval ensure that all surfaces remain off the diagonal. $\square$

The proposition gives existence at every admissible positive radius. It does not give a bound for the kernel as $\delta \downarrow 0$. We next pass from the kernel to its action on harmonic sections; this action has a sphere density that can be estimated separately.

4.4 The transfer and its matrix density

Fix $\delta > 0$ satisfying (4.11), write $r = r_\delta$, and denote the continued kernel by $\mathcal{G}(x, y)$. For a first-system harmonic section u defined near the closed γ -ball centered at y , set

$$S(y; u) = \int_{\partial B_\gamma(y, r(y))} \left[\mathcal{G}(x, y)^* \nabla_{\nu_1}^{A_1} u(x) - (\nabla_{\nu_1, x}^{A_1} \mathcal{G}(x, y))^* u(x) \right] dS_1(x). \quad (4.18)$$

Here ν_1 and dS_1 are the outward unit normal and hypersurface density for the actual metric g_1 . The derivative of the kernel acts on its first fiber index. Consequently the displayed integrand is a section of the second fiber, with the adjoints in the indicated order. For a first harmonic-frame function v , define

$$T_y v = F_2(y)^{-1} S(y; F_1 v). \quad (4.19)$$

Let S_y denote the unit sphere of $(T_y \mathbb{R}^n, \gamma_y)$ and $d\sigma_y$ its rotation-invariant probability measure.

Proposition 4.5 (Matrix harmonic transfer). *Under the assumptions of Proposition 4.4, there is a smooth matrix function $\psi = \psi_\delta$ on the unit sphere bundle over Y° such that*

$$T_y v = \int_{S_y} \psi(y, \omega) v(\exp_y^\gamma(r(y)\omega)) d\sigma_y(\omega). \quad (4.20)$$

For every $y_0 \in Y^\circ$ and every first harmonic-frame function v defined near $\overline{B_\gamma(y_0, r(y_0))}$, the function $y \mapsto T_y v$ is second harmonic-frame harmonic on a neighborhood of y_0 . For each pair (v_1, v_2) of first and second harmonic-frame functions defined on the fixed coordinate neighborhoods and agreeing on the known side,

$$\int_{S_y} \psi(y, \omega) d\sigma_y(\omega) = I, \quad T_y v_1 = v_2(y). \quad (4.21)$$

Proof. The Dirichlet problem on each ball is invertible. In the unitary frame this follows from the positive connection energy: a zero-boundary null solution is parallel and hence zero. Invertibility is preserved by the harmonic-frame change and by positive scalar multiplication of the equation. Its Dirichlet solution operator determines the boundary covariant normal derivative from the boundary value. Transpose this boundary operator onto the smooth matrix coefficient in the first term of (4.18), then multiply by the frame factors and pull the surface density back under $\omega \mapsto \exp_y^\gamma(r(y)\omega)$. This gives the smooth matrix density in (4.20). Smooth dependence of the ball Dirichlet problem on its center and positive radius gives smooth dependence of ψ .

To prove the local harmonicity assertion, fix y_0 and $u = F_1 v$. The kernel is defined on an open neighborhood of the compact sphere over y_0 . Choose a fixed slightly larger surrounding surface inside that neighborhood and inside the domain of u . After restricting the center neighborhood, this surface surrounds all the moving spheres and the intervening collars remain in the common domain. Green's identity on each collar replaces (4.18) by its integral over the fixed surface. Both first-variable equations are homogeneous in the collar. Applying the second-variable operator under the fixed integral now proves that $S(y; u)$ is second-system harmonic: the columns of $\mathcal{G}(x, y)^*$ solve that system. Equation (4.19) proves the assertion in harmonic frames. This argument applies to arbitrary local harmonic inputs; they need not extend throughout the coordinate region.

If $y_n < -1/4$, the kernel is G_1 and the Green representation formula gives $S(y; u) = u(y)$. On that region $F_1 = F_2$, so $T_y v = v(y)$. For a fixed pair (v_1, v_2) , both $T_y v_1$ and $v_2(y)$ solve the second harmonic-frame equation on connected Y° and agree in the lower strip. Unique continuation proves their equality. Finally constant columns are harmonic in both harmonic frames and agree there, so the same argument yields the first identity in (4.21). $\square$

Lemma 4.6 (Bounds where the cutoff varies). *After the choices in Remark 4.2, for every fixed integer m the angular H^m norms of ψ_δ and its first base derivatives on $\text{supp}(d\chi)$ are bounded independently of admissible δ and sufficiently small λ .*

Proof. On the lower cutoff region, T_y is evaluation at the center in the harmonic frame. Normalize its equation to principal matrix $a = \gamma^{-1}$, use a smooth γ -orthonormal frame to pull it back under $z \mapsto \exp_y^\gamma(rz)$, and multiply by r^2 . With center and radius regarded as independent parameters, the resulting Dirichlet systems extend smoothly to $r = 0$, are uniformly elliptic, and have the Euclidean Laplacian as limit; their lower-order coefficients vanish at $r = 0$. The Dirichlet inverses are uniformly bounded, either by their limiting inverse and perturbation or by the positive energy before the change of frame. Elliptic boundary regularity, applied after differentiation in the parameters, shows that the evaluation density at the center has uniform angular smooth bounds and smooth center-radius dependence. The center remains a fixed positive distance from the unit sphere in these coordinates. Substituting the varying radius costs only $|dr|_\gamma = r\beta$, which is uniformly bounded by (4.12) and the lower bound for β .

This evaluation density is the density constructed in Proposition 4.5. Indeed both represent the same functional on traces of functions harmonic near the closed ball. These traces are dense in smooth boundary data. For a prescribed smooth trace $\varphi(\omega)$, solve on the ball of radius $r(1 + \eta)$ with boundary value φ transported along the radial rays. On pulling this problem back to the closed unit ball, its solution z_η converges in every C^k norm to the solution z_0 at radius r , by smooth dependence of the Dirichlet inverse and boundary regularity. Its trace on the original sphere is $z_\eta(\omega/(1 + \eta))$, which converges in $C^k(S_y)$ to $z_0(\omega) = \varphi(\omega)$. The corresponding physical solutions are harmonic near the original closed ball. Thus the two smooth density matrices agree.

On the other cutoff region $t > t_b$, the radii have the positive lower bound (4.10); all their required derivatives are uniformly bounded in δ . Proposition 3.4 gives uniform kernel bounds there. The smooth Dirichlet construction of the density therefore gives the same bounds for ψ_δ and its first base derivatives. Compactness of the cutoff derivative support completes the proof. $\square$

We have constructed a matrix density that reproduces the matched harmonic pairs and has controlled data wherever the cutoff changes. The remaining estimate is a uniform bound on $\chi\psi_\delta$ over the shrinking part of the sphere family.

5 Uniform estimates for the matrix transfer density

The previous section constructs a smooth matrix density on every sphere of positive radius. We prove that its L^2 norm stays bounded when the spheres shrink, provided the two normalized principal matrices differ by $O(r)$. The main analytic input is a scalar coercivity argument from OpenAI [18, Section 5], whose proof we reproduce below. We then compare the matrix equation with that scalar operator. The comparison is of lower differential order; it requires bounded matrix drifts, but no smallness assumption on their difference.

5.1 Moving traces and the matrix equation

Use the reference metric and normalized inverse matrices from (4.1). The harmonic-frame convention (2.11), with the multipliers c_j in (4.2), gives equations whose second-derivative coefficient matrices are positive definite:

$$\mathcal{L}_1 = a^{ij}\partial_i\partial_j + B_1^i\partial_i, \quad \mathcal{L}_2 = Q^{ij}\partial_i\partial_j + B_2^i\partial_i. \quad (5.1)$$

The B_j^i are complex 2×2 matrices acting on columns from the left. There is no zeroth-order term because each column of the defining frame F_j is harmonic. All coefficients have uniform smooth bounds on the fixed coordinate neighborhood for sufficiently small spatial dilations. Suppress the dilation and shrinking parameters for now, and write

$$r = r_\delta, \quad f_* = \log r, \quad \beta = |df_*|_\gamma, \quad f_i = \nabla_i f_*.$$

The profile bounds supplied by Lemma 4.1 have the form

$$\beta \geq \beta_0, \quad \text{Hess}_\gamma f_* \leq -c_0\beta\gamma, \quad |\nabla^j f_*| \leq C_j\beta^j \quad (1 \leq j \leq 8). \quad (5.2)$$

The lower threshold β_0 can be increased by increasing p_* . Throughout this section we impose

$$\sum_{|\alpha| \leq 8} |\partial_y^\alpha h| \leq \varepsilon r, \quad r\beta^2 \leq \varepsilon. \quad (5.3)$$

The first is the bootstrap assumption of Section 4; the second is ensured by the choice of radius amplitude.

Let $S_\gamma Y$ be the tangent unit-sphere bundle with fiber probability measure $d\sigma_y$. Horizontal differentiation ∇ uses parallel transport for γ , including any base tensor indices, and acts componentwise on the fixed harmonic-frame components. The measure $d\sigma_y$ is parallel. Our Hilbert norms use $dV_\gamma(y)d\sigma_y(\omega)$ and the standard Hermitian norm, or the Frobenius norm for matrices.

There are two different transpose operations in the argument. If $\mathcal{C}$ is a sphere operator on columns, its *bilinear row transpose* $\mathcal{C}^t$ is defined by

$$\int_{S_y} (\mathcal{C}^t \psi) V d\sigma_y = \int_{S_y} \psi (\mathcal{C} V) d\sigma_y. \quad (5.4)$$

Here the identity is applied to each row of ψ ; it involves no complex conjugation. In particular, a matrix multiplication M on columns becomes right multiplication $\psi \mapsto \psi M$, whereas a matrix multiplying the output of the transfer map acts on ψ from the left. A star will denote the Hermitian sphere adjoint of a scalar operator, and a dagger its full Hilbert-space adjoint, including the base integration. Real scalar sphere operators have the same bilinear transpose and Hermitian adjoint.

For a column v solving $\mathcal{L}_1 v = 0$ near a closed moving ball, set

$$V(y, \omega) = v(\exp_y^\gamma(r(y)\omega)).$$

Solving the ball Dirichlet problem expresses the boundary derivatives of v in terms of its trace and therefore defines sphere operators $\mathcal{C}_i$ satisfying $\nabla_i V = \mathcal{C}_i V$. For scalar functions write $\Delta_g = \operatorname{div}_g \nabla$. Let $\mathcal{C}_{i,0}$ be the corresponding moving trace operators for $c_1 \Delta_{g_1}$, whose second-derivative coefficient matrix is $a = \gamma^{-1}$. This scalar comparison equation need not have zero coordinate drift.

On each tangent unit ball, let $\mathbf{B}$ be the radial boundary derivative of Euclidean harmonic extension and $\mathbf{A}_i$ its ambient derivatives in a γ -orthonormal frame. On spherical harmonics $\mathcal{H}_k$ of degree k , $\mathbf{B} = k$ and $\mathbf{A}_i : \mathcal{H}_k \rightarrow \mathcal{H}_{k-1}$. The operators are parallel with their tensor indices, and

$$[\mathbf{B}, \mathbf{A}_i^*] = \mathbf{A}_i^*, \quad \sum_i \mathbf{A}_i^* \mathbf{A}_i^* = 0. \quad (5.5)$$

The second identity is the adjoint of the vanishing Laplacian of a harmonic extension. Define

$$\|w(y, \cdot)\|_s = \|(1 + \mathbf{B})^s w(y, \cdot)\|_{L^2(S_y)}.$$

These norms are equivalent to the usual sphere Sobolev norms, uniformly in y . They also apply componentwise to tensors and matrices.

Write Ψ^m for classical angular pseudodifferential operators of order m , scalar or matrix valued as specified. The notation $c(y)\Psi^m$ means that division by $c(y) > 0$ leaves uniformly bounded symbol and operator seminorms in the finitely many orders being used. An allowance β^j for j base derivatives means the corresponding seminorms are bounded by $C_j \beta^j$.

Lemma 5.1 (Trace expansions). *Under (5.2)–(5.3),*

$$\mathcal{C}_{i,0}^* = r^{-1} \mathbf{A}_i^* + f_i \mathbf{B} + E_i + F_i, \quad E_i \in r\Psi^1, \quad F_i \in \Psi^0. \quad (5.6)$$

The normalized remainders $r^{-1}E_i, F_i$ have the allowance $C_j \beta^j$ through six base derivatives. Moreover, on rows,

$$K_i := \mathcal{C}_i^t - \mathcal{C}_{i,0}^* \in \Psi^0, \quad \nabla K_i \in \beta\Psi^0. \quad (5.7)$$

All constants depend on fixed smooth background bounds, not on the lower threshold β_0 once $\beta_0 \geq 1$.

Proof. First regard the radius s as an independent parameter. Pull the first scalar equation back by $z \mapsto \exp_y^\gamma(sz)$ and multiply by s^2 . Its principal coefficients are $\delta^{ij} + O(s^2)$ and its first-order coefficients are $O(s)$. The same statement holds for the system, with scalar principal coefficients times I . Both Dirichlet operators extend smoothly to $s = 0$, where they equal the Euclidean Laplacian. On each fixed Sobolev scale their Dirichlet inverses are uniformly bounded and smooth for sufficiently small s , by elliptic Dirichlet estimates and invertibility at zero.

For these smooth strongly elliptic Dirichlet families on the unit ball, the boundary radial derivative operator is classical of order one. Its principal symbol depends only on the principal

coefficients and the differentiating vector. The boundary symbol construction, obtained by choosing the decaying normal root and solving the successive transport equations, is smooth in (y, s) ; see, for example, Joshi and Lionheart [12, Section 3, Proposition 3.1]. Their factorization is for the Hodge Laplacian; its construction applies to the present Dirichlet systems because their principal symbol is scalar times I , while the transport equations allow matrix coefficients. The true Dirichlet inverse corrects the parametrix by a smoothing operator. Differentiating this inverse on arbitrarily high fixed Sobolev scales bounds every required smooth kernel seminorm of the correction. Thus these assertions hold in the full symbol topology, including smoothing remainders and parameter derivatives.

At $s = 0$ the radial operator is $\mathbf{B}$, and the first s -derivative of its principal symbol vanishes because the principal coefficients have no linear term in s . Taylor expansion in the symbol topology therefore gives, for the scalar equation,

$$\mathbf{B} + s\mathcal{B}_1(y) + s^2\mathcal{B}_2(y, s), \quad \mathcal{B}_1 \in \Psi^0, \quad \mathcal{B}_2 \in \Psi^1. \quad (5.8)$$

The system and scalar radial operators have identical principal symbols for every s and coincide at zero. Their difference is consequently s times a smooth family in Ψ^0 .

Differentiating the center of the exponential map while transporting ω parallel gives a Jacobi field with initial derivative zero. In the unit-ball coordinates its velocity is

$$s^{-1}(e_i + O(s^2));$$

putting $s = r(y)$ adds $f_i\omega$. Tangential derivatives act directly on boundary data and are identical for the scalar and matrix equations. For the radial component use (5.8). The leading terms are $r^{-1}\mathbf{A}_i + f_i\mathbf{B}$; the order-one errors have size $O(r)$ and the order-zero errors are bounded, since $r\beta \leq \varepsilon/\beta$. This proves (5.6) after sphere adjunction. The radial difference $r\Psi^0$, multiplied by the center velocity $O(r^{-1} + \beta)$, proves (5.7). Base derivatives cost at most one factor β each by the profile bounds. Sphere transposition and commutation with the smooth angular connection fields preserve these orders and bounds. $\square$

Write the second system in the reference connection as $\mathcal{L}_2 = Q^{ij}\nabla_i\nabla_j + b_2^i\nabla_i$; equivalently, $b_2^k = B_2^k + Q^{ij}\Gamma_{ij}^k(\gamma)I$ in coordinates. Set $D_i = \nabla_i + \mathcal{C}_i^t$. The matrix density of Proposition 4.5 satisfies

$$P\psi = 0, \quad P = r \left[Q^{ij}D_iD_j + b_2^iD_i \right]. \quad (5.9)$$

Products are covariant products, and b_2^i multiplies $D_i\psi$ from the left. In particular, Q is not differentiated by a base adjoint. Indeed, differentiation of $T_yv = \int \psi V d\sigma_y$ gives $\nabla_i(T_yv) = \int (D_i\psi)V d\sigma_y$ by (5.4). Differentiate once more and apply the second system equation, which holds for every local first-system harmonic v by Proposition 4.5. The resulting expression pairs to zero with all such traces. These traces are dense in smooth boundary data: solve the Dirichlet problems on slightly enlarged balls with transported smooth boundary values and let their radii decrease to the given radius. Smooth parameter dependence gives convergence in every fixed boundary smooth norm. This proves (5.9).

Write the scalar comparison equation on the second side as $c_2\Delta_{g_2} = Q^{ij}\nabla_i\nabla_j + b_{2,0}^i\nabla_i$. Define the scalar operator, acting entrywise on matrices,

$$P_0 = r \left[Q^{ij}D_{i,0}D_{j,0} + b_{2,0}^iD_{i,0} \right], \quad D_{i,0} = \nabla_i + \mathcal{C}_{i,0}^*. \quad (5.10)$$

The next subsections prove coercivity for P_0 . The proof uses only its trace expansion and coefficient bounds; it assumes no scalar transfer density or scalar Green-kernel identity.

5.2 The model operator and its principal perturbations

We separate a degree-raising model operator from the remaining principal and lower-order terms of the moving trace equation. We continue under the hypotheses of Lemma 5.1, including (5.3).

Define

$$D_i^+ = \nabla_i + f_i \mathbf{B}, \quad D_i^- = -\nabla_i + f_i \mathbf{B}, \quad T = \sum_i \mathbf{A}_i^* D_i^+, \quad T^\flat = \sum_i \mathbf{A}_i D_i^-. \quad (5.11)$$

The symbol T here denotes a differential operator on sphere functions; the transfer functional continues to be denoted T_y . Expand (5.10) by (5.6). The identities (5.5) give

$$D_i^+(r^{-1}\mathbf{A}_j^*) = r^{-1}\mathbf{A}_j^* D_i^+, \quad \sum_i \mathbf{A}_i^* \mathbf{A}_i^* = 0.$$

Thus, in ordinary base coordinates and an orthonormal sphere trivialization,

$$P_0 = 2T + \mathcal{U} + \mathcal{J}, \quad (5.12)$$

where $\mathcal{J}$ is a uniformly bounded family in Ψ^1 , with no base derivatives, and

$$\mathcal{U} = \sum_{|\alpha| \leq 2} \mathcal{U}_\alpha(y) \partial_y^\alpha, \quad \mathcal{U}_\alpha \in \varepsilon \beta^{-|\alpha|} \Psi^{2-|\alpha|}. \quad (5.13)$$

Up to five derivatives of these coefficients have the additional allowance $C_j \beta^j$. The second-base-derivative term is precisely

$$r Q^{ij} \partial_{y_i} \partial_{y_j}; \quad (5.14)$$

its coefficients are real scalars independent of the sphere variable.

We detail these bounds. The pure angular second derivative contracted with a vanishes. The remaining contraction is $(h^{ij}/r) \mathbf{A}_i^* \mathbf{A}_j^*$, and belongs to the zero-base-derivative term of (5.13). The mixed correction is $2h^{ij} \mathbf{A}_i^* D_j^+$ and has the stated small bounds. The $D^+ D^+$ term has coefficient r ; its base orders two, one and zero have coefficients $O(r)$, $O(r\beta)$ and $O(r\beta^2)$, respectively. The condition $r\beta^2 \leq \varepsilon$ gives (5.13) for all three. Terms containing E_i obey the same bounds. Terms containing F_i also do so, except for their cross terms with $r^{-1} \mathbf{A}_j^*$, which are bounded angular order-one terms and belong to $\mathcal{J}$. The drift term with $\mathbf{A}_i^*$ belongs to $\mathcal{J}$ as well. Differentiating any of these expressions gives the stated coefficient bounds by (5.2) and (5.3). A horizontal connection written in this trivialization adds a bounded angular first derivative, which preserves the same estimates. The bounds for $\mathcal{J}$ use the fixed background and remain uniform as β_0 is increased.

The decomposition identifies the model operator $2T$ and the second-order perturbation $\mathcal{U}$. To carry the model estimate over to P_0 we shall need more than an ordinary first-derivative estimate: the error forms also contain mixed base and angular derivatives. For a sphere function φ , put

$$\mathcal{N}(\varphi)^2 = \int_Y \left(\|\nabla \varphi\|_0^2 + \beta^2 \|\varphi\|_1^2 + \beta^{-1} \|\nabla \varphi\|_{1/2}^2 + \beta \|\varphi\|_{3/2}^2 \right) dV_\gamma. \quad (5.15)$$

The last two terms of $\mathcal{N}$ place half an angular derivative on $\nabla \varphi$ and three halves on φ . They will bound the remaining error forms left by the comparison with P_0 .

5.3 A comparison of raising and lowering operators

The next estimate supplies these two derivative strengths for the model operator. Spherical harmonics are trace-free symmetric tensors, and comparing their raising and lowering operators is familiar from energy identities in tensor tomography; see, for example, [10, 19]. Here the negative Hessian of the spatial weight supplies the positive term that absorbs the bounded curvature. The particular weighted estimate below is adapted from OpenAI [18, Section 5, weighted raising estimate]. We include the degree calculation because it is the source of the additional half angular derivative used in the perturbation argument.

Lemma 5.2 (Weighted raising estimate). *Let $n \geq 3$, and suppose (5.2) holds on a fixed base region with uniformly bounded reference geometry. There are β_0 and $c > 0$, depending only on that geometry and the constants in (5.2), such that every smooth, compactly base-supported function with zero fiber mean satisfies*

$$\|T\varphi\|^2 - \|T^\flat\varphi\|^2 \geq c\mathcal{N}(\varphi)^2. \quad (5.16)$$

Proof. Both operators change angular degree by exactly one, so it suffices to prove the estimate at one input degree $k \geq 1$. Identify that component with a harmonic homogeneous polynomial $v(z)$ of degree k . Use the Fischer inner product, in which the monomials z^α are orthogonal with squared norm $\alpha!$. Polynomial differentiation $a_i = \partial_{z_i}$ is adjoint to multiplication $M_i = z_i$. On harmonic polynomials of degree k the Fischer squared norm is the normalized squared sphere norm multiplied by

$$N_k = n(n+2) \cdots (n+2k-2), \quad N_0 = 1.$$

For completeness, integration against a standard real Gaussian identifies the Fischer norm on trace-free homogeneous polynomials with their Gaussian L^2 norm; integrating radially then gives the factor N_k . These classical identities are also recorded in [20, Equation (2.1) and Theorem 2.1].

Set $d = k + n/2 - 1$ and use $D_i^\pm = \pm \nabla_i + k f_i$ at this fixed degree. For harmonic polynomials q_i of degree k , the Laplacian of $\sum_i M_i q_i$ is $2 \sum_i a_i q_i$. Since

$$\Delta_z(|z|^2 s) = 4d s \quad (s \in \mathcal{H}_{k-1}),$$

its harmonic projection is $\sum_i M_i q_i - |z|^2 \sum_i a_i q_i / (2d)$. Orthogonality of this projection and its complement shows that its squared norm is

$$\sum_i \|q_i\|^2 + \sum_{i,j} (a_j q_i, a_i q_j) - d^{-1} \left\| \sum_i a_i q_i \right\|^2.$$

Changing from Fischer adjoints to sphere adjoints gives

$$A_i^*|_{\mathcal{H}_k} = \frac{N_{k+1}}{N_k} \Pi_{k+1} M_i,$$

where Π_{k+1} is harmonic projection in homogeneous degree $k+1$. Consequently N_k times the left side of (5.16) is

$$(2d+2) \left[\|D^+ v\|^2 + \sum_{i,j} (a_j D_i^+ v, a_i D_j^+ v) - \frac{1}{d} \left\| \sum_i a_i D_i^+ v \right\|^2 \right] - 2d \left\| \sum_i a_i D_i^- v \right\|^2. \quad (5.17)$$

All norms in this calculation include integration over the base and use the polynomial inner product in the fibers. The factors are $N_{k+1}/N_k = 2d+2$ and $N_k/N_{k-1} = 2d$.

Write $C_{ij} = 2k(\text{Hess}_\gamma f_*)_{ij}$ and

$$C[av] = \int_Y \sum_{i,j} C_{ij} \langle a_i v, a_j v \rangle dV_\gamma.$$

Covariant integrations by parts give

$$\sum_{i,j} (a_j D_i^+ v, a_i D_j^+ v) = \left\| \sum_i a_i D_i^- v \right\|^2 - C[av] + O(k^2) \|v\|^2, \quad (5.18)$$

$$\left\| \sum_i a_i D_i^+ v \right\|^2 \leq k \|D^- v\|^2 - C[av] + O(k^2) \|v\|^2, \quad (5.19)$$

$$\|D^- v\|^2 = \|D^+ v\|^2 + \int_Y \text{tr } C |v|^2 dV_\gamma. \quad (5.20)$$

Here and below a form denoted $O(k^2) \|v\|^2$ has absolute value at most a fixed constant times that quantity. To verify the first two formulas, move a D_j^+ across the pairing and commute $D_j^- D_i^+$ to $D_i^+ D_j^- - C_{ij}$. The reordered term in (5.19) is at most $k \|D^- v\|^2$, because $\sum_i |a_i w|^2 = k |w|^2$ for $w \in \mathcal{H}_k$. The extra commutator is curvature: it acts as a sum of rotations of norm $O(k)$ at degree k , and the two contractions supply one more factor k . This proves the claimed error bounds. Expanding the two squares proves (5.20). For complex functions all the displayed forms are understood through their real parts.

Substituting (5.18) into (5.17) rewrites that expression as

$$\begin{aligned} & (2 + 2/d) \left[d \|D^+ v\|^2 - \left\| \sum_i a_i D_i^+ v \right\|^2 - d C[av] \right] \\ & + 2 \left\| \sum_i a_i D_i^- v \right\|^2 + O(k^3) \|v\|^2. \end{aligned} \quad (5.21)$$

We must control the remaining negative divergence square while retaining both an ordinary derivative estimate and an additional half-order angular derivative estimate. The latter requires a coefficient of size k/β in front of $|D^+ v|^2$, including when k is much larger than β . Choose a small constant $\vartheta > 0$. On the fraction $1 - \vartheta$ of its negative divergence square use (5.19)–(5.20). On the remaining fraction use the pointwise convex combination

$$\begin{aligned} \left| \sum_i a_i D_i^+ v \right|^2 & \leq (1 - \zeta)(k + n - 1) |D^+ v|^2 \\ & + \zeta \left(2 \left| \sum_i a_i D_i^- v \right|^2 + 8k^3 \beta^2 |v|^2 \right), \quad \zeta = c_1/\beta. \end{aligned} \quad (5.22)$$

The fraction $1 - \vartheta$ in the integrated estimates is constant; the y -dependent ζ is used only in this pointwise inequality. The first bound follows from the adjoint multiplication operators, since the contraction $(q_i) \mapsto \sum_i a_i q_i$ has squared norm at most $k + n - 1$ on polynomials of degree k . The second follows from $D_i^+ + D_i^- = 2k f_i$ and $|\sum_i f_i a_i v|^2 \leq k \beta^2 |v|^2$.

Combining these bounds in (5.21), the coefficient retained for $|D^+ v|^2$ is

$$(2 + 2/d)[d - k - \vartheta(n - 1) + \vartheta \zeta(k + n - 1)]. \quad (5.23)$$

Since $d - k = (n - 2)/2$, a sufficiently small fixed ϑ makes this at least a fixed positive constant plus a positive multiple of $c_1 k/\beta$. The coefficient of $|\sum_i a_i D_i^- v|^2$ is nonnegative when β_0 is sufficiently large. The Hessian contribution is exactly

$$-(2 + 2/d) \left[(d - 1 + \vartheta) C[av] + (1 - \vartheta) k \int_Y \text{tr } C |v|^2 dV_\gamma \right]. \quad (5.24)$$

For $k \geq 1$ and $n \geq 3$, both coefficients inside this expression are positive. The Hessian hypothesis bounds it below by $ck^3 \int \beta |v|^2$. This absorbs the curvature error $O(k^3) \|v\|^2$ by increasing β_0 , and absorbs the last term of (5.22) by choosing c_1 sufficiently small. We have therefore retained positive multiples of

$$\|D^+ v\|^2, \quad \int_Y \frac{k}{\beta} |D^+ v|^2 dV_\gamma, \quad k^3 \int_Y \beta |v|^2 dV_\gamma.$$

The first controls both $\|\nabla v\|^2$ and $k^2 \int \beta^2 |v|^2$, since its cross term is $-k \int \Delta_\gamma f_* |v|^2 \geq 0$. For the weighted derivative use the pointwise inequality

$$\frac{k}{\beta} |\nabla v|^2 \leq 2 \frac{k}{\beta} |D^+ v|^2 + 2k^3 \beta |v|^2.$$

No differentiation of β^{-1} is required. Dividing by N_k , and then summing the orthogonal degree components, proves (5.16); the weights $(1+k)^s$ are comparable to k^s because the zero degree is excluded. $\square$

5.4 Keeping the principal perturbations

The raising estimate controls the model operator. The scalar operator (5.10) also contains second base derivatives and a small second-order angular perturbation. We retain these terms in a comparison of squared norms.

Proposition 5.3 (Scalar coercivity). *Fix uniform smooth bounds for the reference geometry and scalar comparison equations, and the constants in (5.2). Suppose P_0 is the scalar operator (5.10), with the scalar trace operators of Lemma 5.1. There exist β_0 sufficiently large, $\varepsilon_0 > 0$ and $C < \infty$, depending only on these fixed bounds, such that (5.3) with $\varepsilon \leq \varepsilon_0$ implies*

$$\mathcal{N}(\varphi) \leq C \|P_0 \varphi\| \quad (5.25)$$

for every smooth, compactly base-supported function φ with zero mean on each sphere. These constants are independent of the shrinking parameter. Only the eight indicated derivatives of the small tensor h are required.

Proof. Let a dagger denote the full adjoint for $dV_\gamma d\sigma$. Covariant integration and the degree shift give

$$Z := T^\flat - T^\dagger = \sum_i f_i \mathbf{A}_i. \quad (5.26)$$

A direct expansion yields

$$\begin{aligned} \|(2T + \mathcal{U})\varphi\|^2 - \|(2T^\flat + \mathcal{U}^\dagger)\varphi\|^2 \\ = 4(\|T\varphi\|^2 - \|T^\flat\varphi\|^2) + \langle \mathcal{E}\varphi, \varphi \rangle, \end{aligned} \quad (5.27)$$

where

$$\mathcal{E} = 2[T^\dagger, \mathcal{U}] + 2[\mathcal{U}^\dagger, T] + [\mathcal{U}^\dagger, \mathcal{U}] - 2(\mathcal{U}Z + (\mathcal{U}Z)^\dagger). \quad (5.28)$$

We will prove

$$|\langle \mathcal{E}\varphi, \varphi \rangle| \leq C\varepsilon \mathcal{N}(\varphi)^2. \quad (5.29)$$

Here are the complete order and weight counts needed for this claim. An operator has count (m, s) if its coefficient at ∂_y^α is angular order $m - |\alpha|$, of size $O(\beta^{s-|\alpha|})$, with the derivative allowances already specified. Adjoints preserve the count, and products add the counts. When a base derivative hits a coefficient it consumes one differential order and

adds a factor β , so it preserves the weight index. A commutator of scalar angular operators loses one angular order: the products of their principal symbols cancel. Hence a scalar commutator loses one in total order while preserving the sum of the weight indices. The counts of $T, T^\dagger, \mathcal{U}, Z$ are, respectively,

$$(2, 1), \quad (2, 1), \quad (2, 0), \quad (1, 1),$$

and every occurrence of $\mathcal{U}$ carries the factor ε . Thus (5.28) has count at most $(3, 1)$ and a factor $O(\varepsilon)$; the quadratic $\mathcal{U}$ term is smaller.

There is a further cancellation that the total count alone does not express: no third base derivative survives. To check it explicitly, use an orthonormal sphere trivialization and write the base measure as $\mu(y) dy$. The sphere probability measure is then fixed. Write

$$\mathcal{U} = c^{ij} \partial_i \partial_j + \mathcal{B}^i \partial_i + \mathcal{C}, \quad c^{ij} = rQ^{ij} = c^{ji} \in \mathbb{R}.$$

The coefficients $\mathcal{B}^i$ and $\mathcal{C}$ are angular operators. Replacing a horizontal derivative by a coordinate derivative plus its angular connection term changes only base orders at most one, so these are all the second-base coefficients. If a star denotes the sphere adjoint, direct integration by parts gives

$$\begin{aligned} \mathcal{U}^\dagger - \mathcal{U} &= \left(2\mu^{-1} \partial_j (\mu c^{ij}) - \mathcal{B}^i - (\mathcal{B}^i)^* \right) \partial_i \\ &\quad + \mu^{-1} \partial_i \partial_j (\mu c^{ij}) - \mu^{-1} \partial_i (\mu (\mathcal{B}^i)^*) + \mathcal{C}^* - \mathcal{C}. \end{aligned}$$

In particular it has base order at most one. In $[T^\dagger, \mathcal{U}]$ and its partner, the possible third-base-derivative coefficient is a commutator of an angular coefficient with rQ^{ij} , which is zero by (5.14). For $[\mathcal{U}^\dagger, \mathcal{U}]$, first write it as $[\mathcal{U}^\dagger - \mathcal{U}, \mathcal{U}]$. The operator $\mathcal{U}^\dagger - \mathcal{U}$ has no second base derivative, since the coefficient in (5.14) is real. Taking adjoints against the smooth base density only produces lower base orders. Its remaining possible third-base-derivative coefficients commute for the same reason. Finally, $\mathcal{U}Z$ already has at most two base derivatives. It follows that $\mathcal{E}$ is a sum of terms of precisely the following three permitted types:

$$\varepsilon \beta^{-1} \Psi^1 \partial_y^2, \quad \varepsilon \Psi^2 \partial_y, \quad \varepsilon \beta \Psi^3. \quad (5.30)$$

This argument also covers lower orders, since $\beta \geq 1$.

For the first type in (5.30), integrate once in the base. The principal form is bounded by $C\varepsilon \int \beta^{-1} \|\partial_y \varphi\|_{1/2}^2$. A derivative of its coefficient costs one factor β and leaves an $\varepsilon \Psi^1 \partial_y$ term. This includes differentiation of the weight itself: $|d\beta| \leq C\beta^2$ implies $|d(\beta^{-1})| \leq C$. The second type has the form bound

$$C\varepsilon \int_Y \|\partial_y \varphi\|_{1/2} \|\varphi\|_{3/2} dV_\gamma \leq C\varepsilon \int_Y \left(\beta^{-1} \|\partial_y \varphi\|_{1/2}^2 + \beta \|\varphi\|_{3/2}^2 \right) dV_\gamma.$$

The third type is bounded by $C\varepsilon \int \beta \|\varphi\|_{3/2}^2$. The lower-order term from differentiating a coefficient satisfies the same bound. Replacing coordinate derivatives by horizontal derivatives adds a bounded angular first derivative, again controlled by these norms. A fixed finite partition of the base region has the same harmless lower-order errors. This proves (5.29).

The constants in the preceding estimates are uniform for all $\beta \geq \beta_0$ once a fixed lower threshold is met. Choose ε_0 small relative to the constant in Lemma 5.2. Dropping the nonnegative second squared norm in (5.27) gives $\mathcal{N}(\varphi) \leq C\|(2T + \mathcal{U})\varphi\|$. Finally,

$$\|\mathcal{J}\varphi\| \leq C\|\varphi\|_{L^2(Y; H^1(S_y))} \leq C\beta_0^{-1} \mathcal{N}(\varphi),$$

so a sufficiently large β_0 absorbs $\mathcal{J}$ and proves (5.25).

All these operations use angular symbol estimates pointwise in the base, and ordinary base differential operators of degree at most two. Formal adjoints, the commutator expansion and the one integration by parts use at most the five base coefficient derivatives specified in (5.13). Forming these coefficients uses at most one base derivative of a trace operator; six trace derivatives therefore suffice. Since its weight dependence is through r and f_i , profile derivatives through order seven suffice, within the eight allowed in (5.2). Differentiating a factor such as h/r five times uses only five derivatives of h , with the remaining factors bounded by the permitted powers of β . Higher angular symbol seminorms depend on the fixed smooth first-metric family: the tensor h is evaluated at the center, so angular differentiation introduces no further center derivatives of it. The scalar drifts have fixed smooth background bounds. Thus the eight smallness derivatives in (5.3) suffice in every fixed dimension; no smallness of all derivatives of h is used. $\square$

5.5 The matrix comparison and the density bound

The scalar calculation is now complete. We use it entrywise before introducing any matrix coefficients into a squared-norm comparison. This order matters: matrix principal symbols need not commute, whereas the preceding commutator argument is scalar.

Proposition 5.4 (System coercivity). *Fix uniform smooth bounds for the harmonic-frame systems and the reference geometry, and the constants in (5.2). There are β_0 sufficiently large, $\varepsilon_0 > 0$, and C , depending only on these bounds, such that (5.3) with $\varepsilon \leq \varepsilon_0$ implies*

$$\mathcal{N}(\varphi) \leq C \|P\varphi\|_{L^2}$$

for every smooth compactly base-supported matrix function φ with zero fiber mean. The constants are independent of the shrinking parameter and of sufficiently small spatial dilations.

Proof. Put $q^i = b_2^i - b_{2,0}^i I$ and use $D_i = D_{i,0} + K_i$ from (5.7). Let L_M denote left multiplication by a matrix M . Expansion, without interchanging any factors, gives the identity

$$\begin{aligned} P - P_0 &= r [Q^{ij} (D_{i,0} K_j + K_i D_{j,0} + K_i K_j) \\ &\quad + L_{q^i} D_{i,0} + L_{b_2^i} K_i]. \end{aligned} \tag{5.31}$$

Thus $D_{i,0} K_j$ denotes composition and includes differentiation of K_j . There is no derivative of q^i or b_2^i in this formula. Left multiplication commutes with the row action of a sphere operator when its matrix depends only on the center, but no such commutation is needed in (5.31).

The trace expansion gives $\mathcal{C}_{i,0}^* \in (r^{-1} + \beta + 1)\Psi^1$. Together with $K_i \in \Psi^0$, $\nabla K_i \in \beta\Psi^0$, and bounded drifts, (5.31) consequently yields

$$\|(P - P_0)\varphi\|_{L^2} \leq C \left(\|\varphi\|_{L^2(Y; H^1(S_y))} + \|r\nabla\varphi\|_{L^2} \right). \tag{5.32}$$

To see the weights explicitly, every term with a derivative of φ in the base is $r\Psi^0\nabla\varphi$. Products of K_i with the leading $r^{-1}\mathbf{A}_j^*$ leave a bounded Ψ^1 term after multiplication by r ; products with $f_j\mathbf{B}$ have coefficient $O(r\beta)$. Differentiating K_j gives $r\beta\Psi^0$. The remaining products have smaller orders and bounded coefficients. Since $r\beta \leq \varepsilon/\beta$, all constants in (5.32) are uniform as the lower threshold β_0 increases.

Apply Proposition 5.3 to the four entries of φ . By the definition of $\mathcal{N}$,

$$\|\varphi\|_{L^2(Y; H^1)} \leq \beta_0^{-1} \mathcal{N}(\varphi), \quad \|r\nabla\varphi\|_{L^2} \leq \varepsilon\beta_0^{-2} \mathcal{N}(\varphi).$$

It follows that

$$\mathcal{N}(\varphi) \leq C_0 \|P\varphi\|_{L^2} + C_0 C(\beta_0^{-1} + \varepsilon \beta_0^{-2}) \mathcal{N}(\varphi).$$

Increase the fixed lower threshold so that the last coefficient is at most $1/2$. This proves the assertion. In the parameter choices for the construction, this increase is made when p_* is chosen, before ε and the radius amplitude are fixed. $\square$

We record two bounds needed to use coercivity on the actual density. First, P_0 is uniformly bounded: the raising part in (5.12) annihilates constants, the zeroth base coefficient of $\mathcal{U}$ is bounded in Ψ^2 , and $\mathcal{J}$ is bounded in Ψ^1 . Equation (5.32) then shows that PI is uniformly bounded in L^2 . Second, for any fixed smooth base cutoff χ ,

$$\|[P, \chi]w\|_{L^2} \leq C_\chi \left(\|w\|_{L^2(\mathcal{O}_\chi; H^1(S_y))} + \|r\nabla w\|_{L^2(S_\gamma \mathcal{O}_\chi)} \right), \quad (5.33)$$

where $\mathcal{O}_\chi$ is any fixed small neighborhood of the supports of its derivatives. Indeed, $[2T, \chi] = 2 \sum_i A_i^*(\nabla_i \chi)$, $\mathcal{U}$ has second base coefficient rQ^{ij} and first base coefficients bounded in Ψ^1 , and $\mathcal{J}$ commutes with χ . The additional first base coefficients of $P - P_0$ are $r\Psi^0$, by (5.31). These observations give exactly (5.33).

Proposition 5.5 (Uniform transfer density). *Choose the fixed geometry as in Section 4, then choose p_* sufficiently large, $\varepsilon > 0$ sufficiently small, and the radius amplitude $R_* > 0$ sufficiently small that $r_\delta \beta_\delta^2 \leq \varepsilon$ for $0 < \delta \leq 1$. For all sufficiently small spatial dilations there is a constant C , independent of the dilation and δ , such that*

$$\|\chi\psi_\delta\|_{L^2(S_\gamma Y)} \leq C \quad (5.34)$$

whenever $\sum_{|\alpha| \leq \delta} |\partial^\alpha h| \leq \varepsilon r_\delta$ on Y . Here χ is the fixed cutoff from (4.9) and ψ_δ is the left-acting matrix density of Proposition 4.5.

Proof. The reproduction identity $\int \psi_\delta d\sigma_y = I$ makes $\varphi = \chi(\psi_\delta - I)$ mean zero on each sphere. It is smooth and compactly supported in the base. Equation (5.9) gives

$$P\varphi = -\chi PI + [P, \chi](\psi_\delta - I).$$

The first term has the uniform bound just established. For the second, (5.33) requires only one angular derivative and $r\nabla\psi_\delta$ on the cutoff derivative region. The uniform bounds of Lemma 4.6 supply these: in the lower region the density is the normalized ball evaluation kernel, and in the other region the radii have a fixed positive lower bound. System coercivity bounds $\mathcal{N}(\varphi)$, hence $\|\varphi\|_{L^2}$, uniformly. Adding the fixed matrix function χI proves (5.34). $\square$

The estimate controls the density of the actual transfer functional. In the next section its first and second Taylor moments will turn this bound into quantitative information about the difference of the two harmonic-frame equations.

6 Matrix moments and extension across the contact point

The transfer density constructed in Section 4 has a uniform L^2 bound whenever the two normalized principal tensors differ by at most a small multiple of the sphere radius. We now improve that comparison from order r to order $r^{3/2}$. This strict improvement will allow the radii to tend to zero while keeping one fixed dilation of the original coordinates.

The scalar argument uses affine and quadratic moments of the transfer density; see [18, Section 6]. In a harmonic frame, the first moments are matrices and need not vanish. We

approximate them by a matrix-valued displacement independent of the shrinking parameter. Its component complementary to the real scalar matrices satisfies a first-order system with conformal Killing principal part. The finite-type property of that system provides the additional power of the radius.

6.1 A finite family of harmonic jets

We use the coordinates and harmonic frames of Proposition 2.5. Before dilation, their matrix coordinate functions satisfy

$$\frac{1}{2} \operatorname{Re} \operatorname{tr} X_j^l(x) = x^l, \quad X_j^l(0) = 0, \quad \partial_i X_j^l(0) = \delta_i^l I, \quad 1 \leq l \leq n.$$

There are also finitely many paired harmonic columns whose value and gradient vanish at the origin and whose first-side Hessians form a basis of the vector-valued trace-free symmetric Hessians there. All these pairs agree on the known side of the contact hypersurface.

For the dilation parameter λ , replace the coordinate matrices by

$$\widehat{X}_j^l(y) = \lambda^{-1} X_j^l(\lambda y), \quad (6.1)$$

and replace each of the additional columns u_j by $\lambda^{-2} u_j(\lambda y)$. Let

$$(v_{1,\nu}, v_{2,\nu}), \quad 1 \leq \nu \leq N,$$

be the resulting list of column pairs, including all columns of the $\widehat{X}_j^l$. The list is finite and fixed independently of the shrinking parameter δ . Taylor's formula at the origin shows that these functions have uniform bounds of every fixed smooth order on the fixed dilation range. To obtain the second-side bounds, note that each paired difference, expressed in the common coordinates, vanishes on an open set accumulating at the origin. Every derivative therefore vanishes at the origin by continuity. Each additional column pair thus has zero value and gradient on both sides, as required for its quadratic rescaling; the coordinate pairs have the common zero value required for linear rescaling. Taylor's formula then gives uniform bounds of every fixed order on both sides.

Recall the harmonic-frame equations, written with positive definite second-derivative coefficients:

$$\mathcal{L}_1 = a^{ij} \partial_i \partial_j + B_1^i \partial_i, \quad \mathcal{L}_2 = Q^{ij} \partial_i \partial_j + B_2^i \partial_i, \quad Q = a + h. \quad (6.2)$$

The tensors a, Q are real and scalar in the fiber index; the B_j^i are complex 2×2 matrices. All coefficients have uniform smooth bounds for small λ . There is no zeroth-order term because the columns of the frames are harmonic. In ordinary coordinates, $B_j^i \rightarrow 0$ smoothly under dilation, whereas $a \rightarrow e^{2Lt_0} I_n$.

Lemma 6.1 (Uniform spanning by the selected jets). *For all sufficiently small λ and every $y \in Y$, the vectors*

$$(\partial_i v_{1,\nu}(y), \partial_i \partial_j v_{1,\nu}(y)), \quad 1 \leq \nu \leq N,$$

span, over $\mathbb{C}$, the space of pairs (p, H) with $p_i \in \mathbb{C}^2$ and $H_{ij} = H_{ji} \in \mathbb{C}^2$ satisfying

$$a^{ij} H_{ij} + B_1^i p_i = 0. \quad (6.3)$$

The coefficients realizing prescribed such data may be chosen by a smooth right inverse, with uniform bounds to every fixed order.

Proof. As $\lambda \rightarrow 0$, the coordinate columns converge smoothly to $y^l e_\alpha$, $1 \leq l \leq n$, $1 \leq \alpha \leq 2$, where e_α is the standard basis of $\mathbb{C}^2$. The additional columns converge to the trace-free quadratic polynomials specified by their Hessians at the origin. At any point y , the latter give arbitrary trace-free Hessians, and the affine columns correct their gradients to any prescribed values. Thus the limiting jets span all (p, H) with $\sum_i H_{ii} = 0$, uniformly for $y \in Y$.

The spaces in (6.3) form a smooth vector bundle of constant rank: contraction with the positive definite tensor a is onto $\mathbb{C}^2$. They can be trivialized by the gradient and the trace-free part of the Hessian, the remaining trace being determined by (6.3). In this trivialization the selected jets give a smooth matrix of constant full row rank at $\lambda = 0$. Compactness of Y preserves this rank, with a positive lower bound on its nonzero singular values, for small λ . If this matrix is T , the right inverse $T^*(TT^*)^{-1}$ has the asserted smoothness and uniform bounds. Each selected column satisfies (6.3), so its range is precisely the stated space. $\square$

6.2 Displacement and weighted remainders

Throughout the next three subsections we fix $0 < \delta \leq 1$ and assume the provisional comparison

$$H_8(y) := \sum_{|\alpha| \leq 8} |\partial^\alpha h(y)| \leq \varepsilon r_\delta(y), \quad y \in Y. \quad (6.4)$$

Write $r = r_\delta$. Since $r_{\delta'} \geq r_\delta$ when $\delta' \geq \delta$, the bound (6.4) also holds with every intermediate radius used to continue from $\delta' = 1$ to $\delta' = \delta$. Proposition 4.5 and Proposition 5.5 give

$$\begin{aligned} v_{2,\nu}(y) &= \int_{S_y} \psi(y, \omega) v_{1,\nu}(\exp_y^\gamma(r(y)\omega)) d\sigma_y(\omega), \\ \int_{S_y} \psi(y, \omega) d\sigma_y(\omega) &= I, \quad \|\chi\psi\|_{L^2(Y \times S_y)} \leq C. \end{aligned} \quad (6.5)$$

The constant is independent of δ and of sufficiently small λ . The measure in the last norm is $dV_\gamma(y)d\sigma_y(\omega)$; on the fixed coordinate range it is uniformly equivalent to the corresponding Euclidean base measure.

Set $x(y, \omega) = \exp_y^\gamma(r(y)\omega)$. The first Taylor coefficients of the transfer formula involve the matrix moments

$$M^i(y) = \int_{S_y} \psi(y, \omega) (x^i(y, \omega) - y^i) d\sigma_y(\omega).$$

These moments depend on δ through the density and the sphere. We need a displacement that remains fixed as the radii shrink. The paired coordinate matrices provide such a choice, and the moment estimate below will show that it differs from M only by a second-order error.

Define matrices $D^i(y)$, $1 \leq i \leq n$, by the linear system

$$\hat{X}_2^l - \hat{X}_1^l = \sum_i D^i \partial_i \hat{X}_1^l, \quad 1 \leq l \leq n. \quad (6.6)$$

The map on the right sends the n matrices D^i to n matrices by right multiplication with the indicated coefficients. It converges uniformly, with derivatives, to the identity map because $\partial_i \hat{X}_1^l \rightarrow \delta_i^l I$. Hence (6.6) defines smooth D^i with uniform smooth bounds. In particular, D is independent of δ . For each selected pair define

$$R_\nu = v_{2,\nu} - v_{1,\nu} - D^i \partial_i v_{1,\nu}. \quad (6.7)$$

These remainders also have uniform smooth bounds and are independent of δ .

Lemma 6.2 (Weighted moment bounds). *Under (6.4),*

$$\|\chi D/r\|_{L^2(Y)} \leq C, \quad \|\chi R_\nu/r^2\|_{L^2(Y)} \leq C, \quad 1 \leq \nu \leq N. \quad (6.8)$$

The constants are uniform in δ and small λ .

Proof. Writing $\rho(y) = \|\psi(y, \cdot)\|_{L^2(S_y)}$, uniform normal coordinates and Cauchy–Schwarz give $|M(y)| \leq Cr(y)\rho(y)$. Taylor expansion in the fixed chart, followed by (6.5), gives

$$v_{2,\nu} - v_{1,\nu} = M^i \partial_i v_{1,\nu} + E_\nu, \quad |E_\nu(y)| \leq Cr(y)^2 \rho(y). \quad (6.9)$$

This calculation uses only the uniform second derivatives of the selected columns and $|x - y| \leq Cr$; the coordinate displacement is a scalar, so no matrix factors have been interchanged.

Apply the same calculation to the coordinate matrices. Subtracting (6.6) shows that the uniformly invertible map defining D sends $D - M$ to an error bounded by $Cr^2\rho$. Therefore

$$|D - M| \leq Cr^2\rho, \quad |D| \leq Cr\rho, \quad |R_\nu| \leq Cr^2\rho.$$

Multiplication by χ and integration prove (6.8). $\square$

The next estimate turns these weighted L^2 bounds into bounds for enough derivatives to compare the equations. No derivative of the shrinking radius is taken in this interpolation step.

Lemma 6.3 (Pointwise interpolation). *Choose*

$$d_* = 10 + \frac{n}{2}, \quad \eta = \frac{1}{2d_*}, \quad p_* > 2d_*, \quad m \geq \lceil 10 + 3d_* \rceil.$$

There is a fixed $\rho_0 > 0$ such that, if $y \in K$ and $r(y) < \rho_0$, then

$$\sum_{|\alpha| \leq 10} |\partial^\alpha D(y)| \leq Cr(y)^{1/2}, \quad \sum_{|\alpha| \leq 10} |\partial^\alpha R_\nu(y)| \leq Cr(y)^{3/2}. \quad (6.10)$$

The threshold and constants are independent of δ and small λ .

Proof. By Lemma 4.1, r^{1/p_*} has a uniform Lipschitz constant. Set $s = r(y)^\eta$. Since $\eta > 1/p_*$, for $r(y)$ sufficiently small every $z \in B(y, s)$ satisfies

$$|r(z)^{1/p_*} - r(y)^{1/p_*}| \leq Cr(y)^\eta \leq \frac{1}{2}r(y)^{1/p_*}.$$

Consequently $cr(y) \leq r(z) \leq Cr(y)$ on this ball, with constants depending only on the fixed p_* . Because $\chi = 1$ on a neighborhood of the compact set K , the ball lies in that neighborhood after a further fixed reduction of ρ_0 . The weighted bounds yield

$$\|D\|_{L^2(B(y,s))} \leq Cr(y), \quad \|R_\nu\|_{L^2(B(y,s))} \leq Cr(y)^2.$$

For any smooth vector- or matrix-valued function U with a bounded C^m norm, Taylor expansion to degree $m - 1$ and equivalence of norms on the finite-dimensional polynomial space give

$$|\partial^\alpha U(y)| \leq Cs^{-|\alpha|-n/2} \|U\|_{L^2(B(y,s))} + C_m s^{m-|\alpha|}, \quad |\alpha| < m. \quad (6.11)$$

Indeed the Taylor remainder has L^2 norm at most $C_m s^{m+n/2}$, and, after rescaling the ball to unit radius, each coefficient of its Taylor polynomial is bounded by that polynomial's L^2 norm. This proves the displayed inequality componentwise.

For $|\alpha| \leq 10$, the first term of (6.11) applied to D is at most $Cr(y)^{1-\eta d_*} = Cr(y)^{1/2}$; for R_ν it is at most $Cr(y)^{2-\eta d_*} = Cr(y)^{3/2}$. The second term is at most $C_m r(y)^{\eta(m-10)} \leq C_m r(y)^{3/2}$, taking $\rho_0 \leq 1$. This proves (6.10). $\square$

6.3 The equation for the displacement

The weighted moments give a preliminary $r^{1/2}$ bound for D . To improve it, we compare the two harmonic equations on the selected columns. This comparison separates the scalar tensor h from the matrix part of D .

For a selected first-side column write $w = v_{1,\nu}$ and $R = R_\nu$. Using $v_{2,\nu} = w + D^k \partial_k w + R$, expand $\mathcal{L}_2 v_{2,\nu} = 0$ and use $\mathcal{L}_1 w = 0$. In the third derivatives use

$$a^{ij} \partial_i \partial_j \partial_k w = -(\partial_k a^{ij}) \partial_i \partial_j w - (\partial_k B_1^i) \partial_i w - B_1^i \partial_i \partial_k w.$$

The resulting identity is

$$\begin{aligned} [h^{ij} I + a^{li} \partial_l D^j + a^{lj} \partial_l D^i + \mathcal{B}^{ij}(D)] \partial_i \partial_j w + \mathcal{Y}^i \partial_i w \\ = -\mathcal{L}_2 R - 2h^{ij} (\partial_i D^k) \partial_j \partial_k w - D^k h^{ij} \partial_i \partial_j \partial_k w, \end{aligned} \quad (6.12)$$

$$\mathcal{B}^{ij}(D) = -D^k \partial_k a^{ij} + \frac{1}{2} (B_2^i D^j + B_2^j D^i - D^j B_1^i - D^i B_1^j), \quad (6.13)$$

$$\mathcal{Y}^k = B_2^k - B_1^k + Q^{ij} \partial_i \partial_j D^k + B_2^i \partial_i D^k - D^l \partial_l B_1^k. \quad (6.14)$$

In particular, $\mathcal{Y}$ is independent of the selected column. The order of the factors in (6.13) records the noncommutativity of the matrix drifts. The drift of $\mathcal{L}_2$ acting on $D^k \partial_k w$ places B_2^i on the left of D^k , while differentiating the first-side equation places B_1^i on its right.

Let Π be the projection of symmetric contravariant spatial tensors onto their γ -trace-free part, acting entrywise on matrices:

$$(\Pi T)^{ij} = T^{ij} - \frac{1}{n} (\gamma_{kl} T^{kl}) a^{ij}.$$

Recall that $\gamma_{ij} h^{ij} = 0$.

Lemma 6.4 (Projected displacement equation). *There is a smooth matrix-valued symmetric tensor $\mathcal{E}$ such that*

$$h^{ij} I + \Pi(a^{li} \partial_l D^j + a^{lj} \partial_l D^i + \mathcal{B}^{ij}(D)) = \mathcal{E}^{ij}. \quad (6.15)$$

For $y \in K$ with $r(y) < \rho_0$,

$$\sum_{|\alpha| \leq 8} |\partial^\alpha \mathcal{E}(y)| \leq C r(y)^{3/2}. \quad (6.16)$$

Proof. Denote the right-hand side of (6.12) for the column $v_{1,\nu}$ by Z_ν . Its derivatives through order eight are bounded by $C r^{3/2}$ on the stated set. For $\mathcal{L}_2 R_\nu$ this uses derivatives of R_ν through order ten in (6.10). Every derivative of either remaining product contains a derivative of h of order at most eight and a derivative of D of order at most nine or eight, respectively. Thus (6.4) and (6.10) give the factor $r r^{1/2}$. The other factors are fixed smooth derivatives of the selected columns.

At each point, Lemma 6.1 permits linear combinations of the identities (6.12) for which the combined gradient is zero and the combined Hessian is any prescribed a -trace-free symmetric covariant tensor with values in $\mathbb{C}^2$. Such Hessians satisfy (6.3) with zero gradient. They annihilate the $\mathcal{Y}$ term and test precisely the γ -trace-free part of the coefficient of the Hessian.

More explicitly, choose a smooth uniformly bounded basis H_s of scalar symmetric Hessians with $a^{ij} (H_s)_{ij} = 0$, and use the right inverse in Lemma 6.1 to realize $H_s e_\alpha$, for $\alpha = 1, 2$, with zero gradient. The resulting combinations of Z_ν give the two columns of the contraction of the projected coefficient with H_s . Inverting this spatial duality defines $\mathcal{E}$ and proves (6.15). The coefficients realizing these prescribed jets are smooth functions of y and give zero combined gradient identically. We form these combinations before differentiating,

so the $\mathcal{Y}$ term is identically zero. The resulting formula expresses $\mathcal{E}$ using only the Z_ν and coefficients with uniformly bounded derivatives. Differentiating that formula through order eight proves (6.16), without introducing derivatives of $\mathcal{Y}$. $\square$

Define the real-linear projection on matrices

$$q(Z) = Z - \frac{1}{2} \operatorname{Re tr}(Z)I, \quad \mathcal{V}_0 = \{Z \in \mathbb{C}^{2 \times 2} : \operatorname{Re tr} Z = 0\}.$$

Write $D^i = d^i I + W^i$, where $d^i \in \mathbb{R}$ and $W^i \in \mathcal{V}_0$. Taking half the real trace in (6.6), and using the exact trace coordinate identity for both sides, gives

$$d^l = -\frac{1}{2} \operatorname{Re tr} \sum_i W^i \partial_i \hat{X}_1^l. \quad (6.17)$$

Thus d is a zero-order real-linear expression in W with uniformly smooth coefficients.

Apply q in the matrix index to (6.15). The term $h^{ij}I$ vanishes, as do all derivative terms involving $d^i I$. Substituting (6.17) in the remaining zero-order terms gives

$$\Pi(a^{li} \partial_l W^j + a^{lj} \partial_l W^i) + \mathcal{B}_0^{ij}(W) = \mathcal{E}_0^{ij}, \quad \mathcal{E}_0 = q(\mathcal{E}). \quad (6.18)$$

Here $\mathcal{B}_0$ is a real-linear zeroth-order operator with uniformly smooth coefficients, and $\mathcal{E}_0$ satisfies (6.16). There are no hidden first derivatives in $\mathcal{B}_0$: differentiation of (6.17) occurs only in the terms proportional to I , which have already been removed by q .

6.4 Finite type and the improved estimate

We prove the finite-type statement needed for (6.18). In Euclidean space, conformal Killing vector fields have degree at most two; see, for example, [7, Section 3]. The calculation below establishes the corresponding injectivity on third derivatives and includes the uniformity needed for the variable-coefficient equation.

Lemma 6.5 (Prolongation of the conformal Killing principal part). *Let $n \geq 3$, let $a = \gamma^{-1}$ be a smooth positive definite tensor on a coordinate domain, and let V be a finite-dimensional real vector space. Suppose that a V -valued vector field W satisfies*

$$\Pi(a^{li} \partial_l W^j + a^{lj} \partial_l W^i) + C^{ij}(W) = E^{ij},$$

where C is a smooth real-linear zeroth-order operator. Every third derivative of W is then a smooth linear combination of derivatives of W through order two and derivatives of E through order two. The coefficient bounds are uniform when a is uniformly elliptic and the indicated background coefficients range in bounded smooth sets.

Proof. Differentiate the equation twice. At a point, its highest-order part is a linear map from the tensor of third derivatives of W to the twice-differentiated trace-free symmetric gradient. We first show that this map is injective. A constant linear change of coordinates sets $a^{ij} = \delta^{ij}$ at the point, and each real component of V can be considered separately.

A tensor in the kernel defines a homogeneous cubic vector field Z whose trace-free symmetric gradient vanishes: that gradient is homogeneous quadratic and all of its second derivatives are zero. Hence

$$\partial_i Z_j + \partial_j Z_i = 2\delta_{ij}\sigma, \quad \sigma = \frac{1}{n} \operatorname{div} Z. \quad (6.19)$$

Differentiating and combining the three permutations of the indices gives

$$\partial_i \partial_j Z_k = \delta_{jk} \partial_i \sigma + \delta_{ik} \partial_j \sigma - \delta_{ij} \partial_k \sigma.$$

Taking divergence in k yields

$$(n-2)\partial_i\partial_j\sigma = -\delta_{ij}\Delta\sigma.$$

Its trace gives $(2n-2)\Delta\sigma = 0$; since $n \geq 3$, the Hessian of σ is zero. The preceding formula therefore gives $\partial_i\partial_j\partial_l Z_k = 0$, so the homogeneous cubic Z vanishes. This proves injectivity.

Let T_a denote this injective map on third-derivative tensors, using fixed Euclidean inner products on the finite-dimensional tensor spaces. It has the smooth left inverse $(T_a^*T_a)^{-1}T_a^*$. Uniform ellipticity and compactness of bounded coefficient ranges give uniform bounds for this inverse. All other terms in the twice-differentiated equation contain at most two derivatives of W or of E . Applying the left inverse proves the claim, including the smooth coefficient bounds after further fixed numbers of differentiations. $\square$

Proposition 6.6 (Strict improvement of the principal tensors). *With the fixed choices in Lemma 6.3, there are constants C and $\rho_0 > 0$, uniform for sufficiently small λ , such that (6.4) implies*

$$H_8(y) \leq Cr_\delta(y)^{3/2} \quad \text{if } y \in K \text{ and } r_\delta(y) < \rho_0. \quad (6.20)$$

On the same set, the derivatives of D through order nine are bounded by $Cr_\delta^{3/2}$.

Proof. Apply Lemma 6.5 to (6.18) with $V = \mathcal{V}_0$. For each multi-index α of order three, it gives an identity of the form

$$\partial^\alpha W = \sum_{|\zeta| \leq 2} C_{\alpha\zeta} \partial^\zeta W + \sum_{|\zeta| \leq 2} F_{\alpha\zeta} \partial^\zeta \mathcal{E}_0, \quad (6.21)$$

with uniformly smooth coefficient maps.

Fix $y \in K$ with $r(y) < \rho_0$. By the core geometry of Section 4.2, the vertical segment

$$z(s) = (y', s), \quad -\frac{1}{2} \leq s \leq y_n,$$

stays in K , and $r(z(s)) \leq r(y)$. At its initial point, D and all its derivatives vanish: the paired coordinate matrices agree on an open neighborhood in the known lower region. The same holds for W .

Let $\mathcal{W}(s)$ consist of all coordinate derivatives of W through order two at $z(s)$. Differentiating this vector with respect to s either gives another one of its components or a third derivative, to which (6.21) applies. Thus

$$\mathcal{W}'(s) = A(s)\mathcal{W}(s) + F(s), \quad \mathcal{W}(-\tfrac{1}{2}) = 0, \quad |A(s)| \leq C, \quad |F(s)| \leq Cr(y)^{3/2}.$$

The bound for F follows from (6.16), since the entire segment has radius below ρ_0 . Its length is uniformly bounded, so the elementary integral form of Gronwall's inequality gives

$$\sum_{|\alpha| \leq 2} |\partial^\alpha W(y)| \leq Cr(y)^{3/2}.$$

This argument holds at every point of the stated set. Differentiating (6.21) $k-3$ times, for $3 \leq k \leq 9$, expresses the derivatives of W of order k in terms of its derivatives through order $k-1$ and derivatives of $\mathcal{E}_0$ through order $k-1$. Induction and (6.16) therefore give the same bound through order nine. Notice that the last step uses precisely eight derivatives of $\mathcal{E}_0$.

Equation (6.17) now gives the same estimates for d and hence for D . Finally, differentiate (6.15) through order eight. Its right-hand side uses derivatives of D through order nine and of $\mathcal{E}$ through order eight, all bounded by $Cr^{3/2}$. This proves (6.20). $\square$

6.5 One dilation for all shrinking radii

We have proved the strict improvement assuming the provisional bound. We now remove that assumption for all positive shrinking parameters. The order of choices matters: the threshold for the improvement must be fixed before the final choice of the dilation.

Proposition 6.7 (Local equality of the normalized equations). *In the contact coordinates and harmonic frames of Proposition 2.5, there is a neighborhood of the origin on which the two metrics are smoothly conformal and the harmonic-frame equations, multiplied by the positive scalars making their principal tensors equal, coincide. All the finitely many selected pairs of harmonic-frame columns agree on that neighborhood.*

Proof. First fix the reference geometry and uniform smooth bounds for all sufficiently small dilations. Choose p_* large enough for the sphere geometry, the angular estimate, and Lemma 6.3. Next choose ε sufficiently small and R_* sufficiently small to satisfy those results, including $r\beta^2 \leq \varepsilon$. As explained in Remark 4.2, every fixed-separation range needed for the continuation and cutoff bounds is then fixed before λ . Proposition 5.5 and all estimates of this section have constants uniform for sufficiently small λ .

Choose $0 < \rho_* < \rho_0$, with $\rho_* \leq 1$, so small that $C\rho_*^{1/2} \leq \varepsilon/2$ in (6.20). Thus, wherever that estimate applies and $r_\delta < \rho_*$, it improves (6.4) to $H_8 \leq \varepsilon r_\delta/2$.

As $\lambda \rightarrow 0$, $h \rightarrow 0$ in $C^8(Y)$. The numbers

$$\rho_*, \quad \min_Y r_1, \quad R_*(2t_b)^{p_*}$$

are positive and fixed independently of λ . We can therefore choose one sufficiently small λ such that

$$\sup_Y H_8 \leq \frac{\varepsilon}{2} \min \{ \rho_*, \min_Y r_1, R_*(2t_b)^{p_*} \}. \quad (6.22)$$

We also make the known lower region contain $\{y_n < -1/4\}$ on the fixed ranges, as in the sphere construction. Fix this λ for the rest of the proof.

Let

$$\mathcal{I} = \{ \delta \in (0, 1] : H_8(y) \leq \varepsilon r_\delta(y) \text{ for every } y \in Y \}.$$

It contains 1 by (6.22). For any $\delta \in \mathcal{I}$, the preceding estimates improve the inequality to $H_8 \leq \varepsilon r_\delta/2$ throughout Y . Indeed, Proposition 6.6 handles the points in K with $r_\delta < \rho_*$. Equation (6.22) handles all points with $r_\delta \geq \rho_*$. At a point outside K , either $y_n < -1/2$, where equality of the coefficients is already known, or $t > 2t_b$. In the latter case $\ell_\delta(t) \geq t$, so $r_\delta \geq R_*(2t_b)^{p_*}$ and (6.22) again applies.

The set $\mathcal{I}$ is relatively closed in $(0, 1]$ by continuity and compactness of Y . It is also relatively open: at any positive δ , the radius has a positive minimum on Y , so the strict factor $1/2$ just proved persists for nearby parameter values in the weaker defining inequality. Since $(0, 1]$ is connected, $\mathcal{I} = (0, 1]$.

On the fixed open set $K^\circ \cap \{t < 0\}$, which contains the origin, $r_\delta \rightarrow 0$ as $\delta \rightarrow 0$. The coefficients h, D, R_ν are independent of δ . The provisional bound and (6.10) therefore imply

$$h = 0, \quad D = 0, \quad R_\nu = 0$$

there. In particular all selected pairs agree. Subtracting their equations now gives

$$(B_2^i - B_1^i) \partial_i v_{1,\nu} = 0 \quad (1 \leq \nu \leq N).$$

The gradient projection of the space (6.3) is all of $(\mathbb{C}^2)^n$: for any gradient one can choose the trace of the Hessian to satisfy that constraint. Lemma 6.1 therefore implies $B_1 = B_2$.

Together with $Q = a$, this proves equality of the normalized harmonic-frame operators. The definition of Q shows that the two inverse metrics are positive scalar multiples, so the metrics are smoothly conformal. Undoing the fixed dilation proves the stated conclusion in the original contact coordinates. $\square$

The remaining conformal factor and the metric carried by the harmonic frames are addressed next. The local conclusion just proved supplies the equality of equations needed to recover both the Riemannian metric and a unitary connection identification.

7 Exact identification and the measured boundary

Proposition 6.7 identifies the metric up to conformal scale and identifies the normalized equations in harmonic frames near a contact point. We first show why, for a connection Laplacian without an additional potential, this is already an exact local isometry with a unitary connection identification. We then return to the maximal match of Section 2 and prove that it extends over the whole manifold and fixes every part of the measured boundary.

7.1 Removing the conformal factor

Lemma 7.1 (Exact local identification). *In the contact coordinates and unitary frames of Proposition 2.5, suppose that on a connected neighborhood Ω of zero the metrics satisfy $g_2 = e^{2\omega}g_1$, and that the equations in the harmonic frames F_j , after scalar normalization to the same principal part, have equal first-order coefficients. Then*

$$g_2 = g_1, \quad J_{\text{loc}} = F_1 F_2^{-1} \in U(2), \quad d_{A_1}(J_{\text{loc}}u) = J_{\text{loc}}d_{A_2}u \quad \text{on } \Omega.$$

Moreover this identification agrees with the old match on the known side and matches all corresponding source evaluations throughout Ω .

Proof. Let

$$C_j = F_j^{-1}(dF_j + A_j F_j).$$

These are the connection matrices in the harmonic frames; they need not be skew-Hermitian in the standard matrix inner product. The negative of the actual connection Laplacian, expressed in such a frame, is

$$\Delta_{g_j, C_j} = g_j^{il}((\partial_i + C_{j,i})(\partial_l + C_{j,l}) - \Gamma_{il}^k(g_j)(\partial_k + C_{j,k})).$$

Its zeroth-order term is zero, because the columns of F_j are harmonic. The common-principal-part hypothesis therefore says

$$e^{2\omega} \Delta_{g_2, C_2} = \Delta_{g_1, C_1}. \tag{7.1}$$

Indeed their second-order coefficients agree after the indicated conformal normalization, their first-order coefficients agree by assumption, and both zeroth-order coefficients vanish.

Put $\phi = (n-2)\omega/2$. The conformal change of the Christoffel symbols gives

$$e^{2\omega} \Delta_{g_2, C_2} = \Delta_{g_1, C_2} + 2\langle d\phi, \partial + C_2 \rangle_{g_1}.$$

Comparison of first-order terms with (7.1) yields $C_2 = C_1 - d\phi I$. Substitution, with matrix multiplication in its displayed order, gives

$$\Delta_{g_1, C_1 - d\phi I} + 2\langle d\phi, \partial + C_1 - d\phi I \rangle_{g_1} = \Delta_{g_1, C_1} - (\Delta_{g_1} \phi + |d\phi|_{g_1}^2)I.$$

Here $\Delta_{g_1} = \operatorname{div}_{g_1} \nabla_{g_1}$. The zeroth-order terms in (7.1) consequently imply

$$\Delta_{g_1} \phi + |d\phi|_{g_1}^2 = 0, \quad \Delta_{g_1}(e^\phi) = 0.$$

On the known open side the metrics already agree, so $e^\phi = 1$. Unique continuation on Ω gives $e^\phi = 1$ everywhere. Since $n \geq 3$ and ω is real, $\omega = 0$, and hence $C_1 = C_2$.

The last equality is equivalent to $d_{A_1}(F_1 F_2^{-1} u) = F_1 F_2^{-1} d_{A_2} u$. Although the harmonic frames are not unitary, $J_{\text{loc}} = F_1 F_2^{-1}$ is. To see this, parallel transport the intertwining identity along any path starting in the known side. Both connection transports are unitary, and J_{loc} is the old unitary identification at the initial point. It remains unitary along the path; connectedness of Ω proves the assertion everywhere.

For an arbitrary source $f \in C_c^\infty(U_0; \mathbb{C}^2)$, both $w_{1,f}$ and $J_{\text{loc}} w_{2,f}$, in the common coordinate domain, are solutions of the first homogeneous connection equation. The charts avoid $\overline{U_0}$. Their difference vanishes on the known side, so unique continuation makes it zero on Ω . Thus all evaluation identities hold, not only those for the finite family used in the local argument. On an overlap with the old match, the two proposed first-side images of a point, together with their invertible fiber maps, would impose an invertible linear relation between the two first-side evaluation maps. Joint surjectivity in Lemma 2.3 forces the image points to coincide. Surjectivity at the second-side point then forces the fiber maps to coincide as well. Thus the local identification agrees with the old match on every overlap. $\square$

7.2 The match fills both interiors

The source-evaluation argument below follows the embedding architecture of [15, Section 6.1]. The maximal matching, boundary completion, and recovery on the whole measured patch adapt [18, Section 7]; we give the details for the unitary bundle identification as well as the base map.

Proposition 7.2 (Global interior identification). *The maximal match of Lemma 2.4 is defined on N_2° , and its base map is an onto isometry $\Phi : N_2^\circ \rightarrow N_1^\circ$.*

Proof. If its domain had an interior frontier, Proposition 2.5 and the dilation in Section 2.5 would supply the local data used in Sections 4–6. Proposition 6.7 gives conformally equal metrics and equal normalized harmonic-frame equations on a neighborhood of the contact point in the original charts. Apply Lemma 7.1. Coordinate identity and J_{loc} extend the old match to a neighborhood containing the frontier point. The concluding assertion of Lemma 7.1 ensures agreement on every overlap. This contradicts maximality. The domain, already nonempty and open, has no relative frontier in the connected interior, and is therefore all of N_2° .

The image is open because Φ is a local isometry. To see that it is relatively closed, suppose $\Phi(p_k) \rightarrow q \in N_1^\circ$. Compactness gives a subsequence with $p_k \rightarrow p \in N_2$ and $J(p_k) \rightarrow J_0 \in U(2)$. If $p \in \partial N_2$, continuity of the evaluations and (2.6) would give $I_1(q) = J_0 I_2(p) = 0$, contrary to Lemma 2.3. Thus p is interior, and continuity gives $\Phi(p) = q$. Connectedness of N_1° makes the image all of that interior. Injectivity was proved in Lemma 2.4; the inverse is smooth because the map is a bijective local diffeomorphism. $\square$

7.3 Boundary extension and the original gauge

Proposition 7.3 (Completion and the measured patch). *The interior maps (Φ, J) extend smoothly to the extended manifolds, with $\Phi : N_2 \rightarrow N_1$ a diffeomorphism and J a unitary bundle map intertwining the connections. They preserve the original copies of M and satisfy*

$$\Phi|_\Gamma = \operatorname{Id}, \quad J|_\Gamma = I.$$

Proof. An onto interior isometry preserves lengths of curves in both directions and therefore preserves intrinsic distance. The metric completion of the interior of a compact smooth Riemannian manifold with boundary is the manifold with its length metric. In particular, a curve touching the boundary can be pushed slightly into an inward collar with arbitrarily small change of length; uniform local metric comparisons identify the resulting completion with the original compact topology. Thus Φ and its inverse extend to mutually inverse homeomorphisms of N_2, N_1 , mapping boundary to boundary.

The extensions are smooth. In a sufficiently thin uniform collar, the distance s_j to ∂N_j is smooth and its gradient generates inward unit normal flow. The isometry preserves distance to boundary, so $s_1 \circ \Phi = s_2$, and in the interior it intertwines these gradient fields. On a fixed small level $s_2 = s_0 > 0$, the map is already smooth. For $0 \leq s \leq s_0$, express it using that smooth level map and the smooth normal flows on the two sides. This gives a smooth extension to $s = 0$; the inverse has the same construction. Parallel transport for A_1, A_2 along these normal curves similarly extends J smoothly, retaining unitarity and the connection intertwining identity.

The maps are identity on the open cap, hence on its closure by continuity. With closure taken in N_j , the original interiors satisfy

$$M_j^\circ = N_j^\circ \setminus \overline{E}.$$

Bijectivity and identity on $\overline{E}$ preserve these complements, and hence their closures, the original copies of M . The restricted diffeomorphism fixes the attachment patch $P = \{b > 0\}$, and $J = I$ there.

We must recover the boundary identity on every part of Γ , which may have more than one component. For $f \in C_c^\infty(\Gamma; \mathbb{C}^2)$, let u_j now denote the harmonic solutions on the original manifolds with boundary value f . On the second manifold define $\tilde{u}_1 = J^{-1}(u_1 \circ \Phi)$. The metric and connection identities make $\tilde{u}_1$ second-harmonic. On P , it has the same boundary value as u_2 . It also has the same covariant normal derivative there. Indeed equality of the measured forms, tested against arbitrary functions supported in P , gives the equality of smooth densities

$$(\nabla_{\nu_1}^{A_1} u_1) dS_{g_1} = (\nabla_{\nu_2}^{A_2} u_2) dS_{g_2} \quad \text{on } P.$$

The densities agree by Lemma 2.1, the isometry transports the outward normal, and the bundle map equals I on P . Thus $\tilde{u}_1 - u_2$ has zero Cauchy data there. Boundary and interior unique continuation give $\tilde{u}_1 = u_2$ throughout M .

Taking boundary traces yields

$$J(p)f(p) = f(\Phi(p)) \quad (p \in \partial M, f \in C_c^\infty(\Gamma; \mathbb{C}^2)). \quad (7.2)$$

If $p \in \Gamma$ and $\Phi(p) \neq p$, choose f nonzero at p , supported in a small neighborhood of p avoiding $\Phi(p)$. Equation (7.2) contradicts invertibility of $J(p)$. Hence $\Phi(p) = p$ on Γ . Arbitrary vector values of $f(p)$ then give $J(p) = I$ there. $\square$

Proof of Theorem 1.1. The boundary-identity gauges V_j of Section 2 put the connections in normal gauge. Lemma 2.1 and Proposition 2.2 produce common exterior Green data. Lemmas 2.3 and 2.4 define the maximal match, and Proposition 2.5 supplies the local contact data wherever an interior frontier is present. Proposition 6.7, followed by Lemma 7.1, removes that frontier. Propositions 7.2 and 7.3 give a smooth global metric and connection identification fixing the entire measured patch.

To distinguish the original connections in this final step, write A_j^{orig} for them, so the connections used in the proof were

$$A_j = V_j^{-1} A_j^{\text{orig}} V_j + V_j^{-1} dV_j.$$

In the original trivializations the bundle map from second to first fiber is

$$U = (V_1 \circ \Phi)JV_2^{-1}.$$

It is smooth and unitary, and equals I on Γ , since both original gauges equal I on the entire boundary. The intertwining identity is $d_{\Phi^*A_1^{\text{orig}}}(Uu) = Ud_{A_2^{\text{orig}}}u$. Expanding it gives

$$A_2^{\text{orig}} = U^{-1}(\Phi^*A_1^{\text{orig}})U + U^{-1}dU, \quad g_2 = \Phi^*g_1.$$

These are the asserted identities. The argument uses densities and a single nonempty cap; it requires neither orientability nor connectedness of the boundary or of Γ . $\square$

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